Vector Calculus

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(adapted from Fieller [1], Section 7.2)

This section covers derivatives of functions of vectors and matrices. Its linear algebra prerequisites, such as vectors, matrices, transposes, dot products, and quadratic forms, are covered in Linear Algebra.

Let $\underset{\sim}{x}$ and $\underset{\sim}{\beta}$ be column vectors of length p (see column vector and dot product).

Definition 0.1 (Partial derivative)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ and $\underset{\sim}{x} = (x_1, \dots, x_p)^\top \in \mathbb{R}^p$. The **partial derivative** [2] of f with respect to x_j at $\underset{\sim}{x}$ is

$$\frac{\partial}{\partial x_j} f(\underset{\sim}{x}) \stackrel{\text{def}}{=} \lim_{h \to 0} \frac{f(x_1, \dots, x_j + h, \dots, x_p) - f(x_1, \dots, x_p)}{h},$$

when this limit exists. It is the derivative at x_j of the one-variable function $t \mapsto f(x_1, \dots, t, \dots, x_p)$, which varies the jth coordinate and holds the other coordinates fixed.

Example 0.1 (Partial derivatives of $x_1^2 x_2 + 3x_2$)**.** Let $f(x_1, x_2) = x_1^2 x_2 + 3x_2$. At $\underset{\sim}{x} = (1, 2)^\top$,

$$\begin{aligned} f(1, 2) &= 1 \cdot 2 + 6 \\ &= 8, \end{aligned}$$

and for $h \neq 0$

$$
\begin{aligned}
\frac{f(1+h,2)-f(1,2)}{h} &= \frac{(1+h)^2\cdot 2+6-8}{h} && \text{(substitute into } f\text{)}\\
&= \frac{(1+h)^2\cdot 2-2}{h} && (6-8=-2)\\
&= \frac{(1+2h+h^2)\cdot 2-2}{h} && \text{(expand } (1+h)^2\text{)}\\
&= \frac{2+4h+2h^2-2}{h} && \text{(distributive law)}\\
&= \frac{4h+2h^2+2-2}{h} && \text{(reorder the terms)}\\
&= \frac{4h+2h^2}{h} && (2-2=0)\\
&= 4+2h, && \text{(divide by } h\neq 0\text{)}
\end{aligned}
$$

which tends to 4 as $h \to 0$, so $\frac{\partial}{\partial x_1} f(1,2) = 4$. Holding x_2 fixed and differentiating in x_1 gives the same answer: $\frac{\partial}{\partial x_1} f(\underset{\sim}{x}) = 2x_1x_2$, which is $2\cdot 1\cdot 2 = 4$ at $(1,2)^\top$. Holding x_1 fixed instead gives $\frac{\partial}{\partial x_2} f(\underset{\sim}{x}) = x_1^2 + 3$, which is $1+3=4$ at $(1,2)^\top$.

Exercise 0.1 (First-order partial derivatives)**.** Adapted from Miller [3], Question 1.1.42. Let $f(x,y) = x^2y + e^x + \sin(xy)$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution

Solution 0.1. To compute $\frac{\partial f}{\partial x}$, treat y as a constant:

$$\frac{\partial f}{\partial x} = 2xy + e^x + y\cos(xy)$$

To compute $\frac{\partial f}{\partial y}$, treat x as a constant:

$$
\begin{aligned}
\frac{\partial f}{\partial y} &= x^2 + 0 + x\cos(xy)\\
&= x^2 + x\cos(xy)
\end{aligned}
$$

Exercise 0.2 (Partial derivatives of the normal density with respect to parameters)**.** Adapted from Miller [3], Question 1.1.43.
Let

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

Find $\frac{\partial f}{\partial \mu}$ and $\frac{\partial f}{\partial \sigma}$.

Solution

Solution 0.2. **1. Derivative with respect to μ:** Treat σ as constant. Differentiating the exponential factor via the chain rule:

$$\begin{aligned}\frac{\partial f}{\partial \mu} &= \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}\cdot\left[-\frac{2(x-\mu)(-1)}{2\sigma^2}\right] \\ &= \frac{x-\mu}{\sigma^2}f(x;\mu,\sigma)\end{aligned}$$

2. Derivative with respect to σ: Write $f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}}\sigma^{-1}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$ and apply the product rule:

$$\begin{aligned}\frac{\partial f}{\partial \sigma} &= -\frac{1}{\sqrt{2\pi}}\sigma^{-2}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} + \frac{1}{\sqrt{2\pi}}\sigma^{-1}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}\cdot\left[\frac{(x-\mu)^2}{\sigma^3}\right] \\ &= \left[-\frac{1}{\sigma} + \frac{(x-\mu)^2}{\sigma^3}\right]f(x;\mu,\sigma)\end{aligned}$$

Remark: Setting $\frac{\partial f}{\partial \mu} = 0$ and $\frac{\partial f}{\partial \sigma} = 0$ for a sample of independent observations leads directly to the values that make the sample most probable, which statistics calls the maximum likelihood estimates: $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$.

Theorem 0.1 (Partial derivatives of the normal density with respect to its parameters)**.** *Let $\mu \in \mathbb{R}$ and $\sigma > 0$, and let*

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

Then for every $x \in \mathbb{R}$,

$$\begin{aligned}\frac{\partial f}{\partial \mu} &= \frac{x-\mu}{\sigma^2}f(x;\mu,\sigma) \\ \frac{\partial f}{\partial \sigma} &= \left[-\frac{1}{\sigma} + \frac{(x-\mu)^2}{\sigma^3}\right]f(x;\mu,\sigma)\end{aligned}$$

Proof

Proof. This is the solution to Exercise 0.2, worked in Solution 0.2. The argument has two steps.

1. For $\frac{\partial f}{\partial \mu}$, treat σ as constant. The prefactor is constant, so the chain rule on the exponential gives $\frac{\partial f}{\partial \mu} = f(x;\mu,\sigma) \cdot \left[-\frac{2(x-\mu)(-1)}{2\sigma^2}\right] = \frac{x-\mu}{\sigma^2} f(x;\mu,\sigma)$.
2. For $\frac{\partial f}{\partial \sigma}$, write $f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}}\sigma^{-1}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$. The product rule gives one term from differentiating σ^{-1}, namely $-\sigma^{-1} f(x;\mu,\sigma)$, and one from the chain rule on the exponential, namely $\frac{(x-\mu)^2}{\sigma^3} f(x;\mu,\sigma)$. Adding them gives $\left[-\frac{1}{\sigma} + \frac{(x-\mu)^2}{\sigma^3}\right] f(x;\mu,\sigma)$.

□

Exercise 0.3 (Partial derivatives with exponential of quadratic form)**.** Adapted from Miller [3], Question 1.1.44.
Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for

$$f(x,y) = x\exp\{x^2+y^2\}$$

Solution

Solution 0.3. With respect to x, apply the product rule:

$$\begin{aligned}\frac{\partial f}{\partial x} &= (1)\exp\{x^2+y^2\} + x\cdot[2x\exp\{x^2+y^2\}] \\ &= (1+2x^2)\exp\{x^2+y^2\}\end{aligned}$$

With respect to y, treat x as constant:

$$\begin{aligned}\frac{\partial f}{\partial y} &= x\cdot[2y\exp\{x^2+y^2\}] \\ &= 2xy\exp\{x^2+y^2\}\end{aligned}$$

Exercise 0.4 (Using symmetry to compute partial derivatives)**.** Adapted from Miller [3], Question 1.1.45.
Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for

$$f(x,y) = e^{xy} - \log(x^2+y^2)$$

on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Solution

Solution 0.4. Holding y constant, differentiate with respect to x:

$$\frac{\partial f}{\partial x} = ye^{xy} - \frac{2x}{x^2+y^2}$$

Notice that the function is symmetric in x and y: $f(x,y) = f(y,x)$. Interchanging x and y gives $\frac{\partial f}{\partial y}$ immediately:

$$\frac{\partial f}{\partial y} = xe^{xy} - \frac{2y}{x^2+y^2}$$

Remark: Recognizing symmetry in multi-variable functions eliminates redundant calculations.

Definition 0.2 (Vector derivative, gradient)**.** If $f(\underset{\sim}{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\underset{\sim}{\beta}$, such as $f(\underset{\sim}{\beta}) = \underset{\sim}{x} \cdot \underset{\sim}{\beta}$, then its **vector derivative** is the column vector of its partial derivatives (Definition 0.1):

$$\frac{\partial}{\partial \underset{\sim}{\beta}} f(\underset{\sim}{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\underset{\sim}{\beta}) \\ \frac{\partial}{\partial \beta_2} f(\underset{\sim}{\beta}) \\ \vdots \\ \frac{\partial}{\partial \beta_p} f(\underset{\sim}{\beta}) \end{bmatrix}$$

The vector derivative is also called the **gradient** [4] of f with respect to $\underset{\sim}{\beta}$, written $\nabla_{\underset{\sim}{\beta}} f(\underset{\sim}{\beta})$.

Example 0.2 (A vector derivative)**.** Let $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$ and $f(\underset{\sim}{\beta}) = 3\beta_1 + 5\beta_2^2$. Then

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} f(\underset{\sim}{\beta})}_{2\times 1} = \begin{bmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1 + 5\beta_2^2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1 + 5\beta_2^2) \end{bmatrix}$$
$$= \begin{bmatrix} 3 \\ 10\beta_2 \end{bmatrix},$$

and at $\underset{\sim}{\beta} = (1,2)^\top$ the vector derivative is $(3,20)^\top$.

Video lecture

Hutchinson's Gradients Refresher (17 min) covers gradients and partial derivatives, the ideas behind this section [5]. The login for the video site is posted on Canvas.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on partial derivatives:

- What Partial Derivatives Are (Hands-on Introduction)
- Partial Derivative Exercises
- Advanced Partial Derivatives
- Advanced Partial-Derivative Exercises
- Partial Derivative Notation

Checking a gradient by its shape

The gradient always has the same shape as the input it is taken with respect to, because it holds exactly one partial derivative per component of that input:

Table 1: The shape of a gradient follows the shape of what it differentiates with respect to.

f is a function of	its gradient is
a single number	a single number — the ordinary derivative
a vector in $\mathbb{R}^p$	a vector in $\mathbb{R}^p$
a matrix in $\mathbb{R}^{m\times n}$	a matrix in $\mathbb{R}^{m\times n}$

The table is the cheapest check there is on a gradient just worked out by hand: if the shapes disagree, something went wrong. It is the shape rule of matrix multiplication again, applied to calculus.

Definition 0.1 (Linear approximation of a function of a vector, first-order Taylor approximation of a function of a vector)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have a gradient at β. The **linear approximation** of f at β [6] (also called the **first-order Taylor approximation** or the **first-order Taylor expansion**) is the function of a step $\underset{\sim}{\delta} \in \mathbb{R}^p$

$$\hat{f}_{\underset{\sim}{\beta}}(\underset{\sim}{\delta}) \stackrel{\text{def}}{=} f(\underset{\sim}{\beta}) + \nabla f(\underset{\sim}{\beta}) \cdot \underset{\sim}{\delta}. \tag{1}$$

Example 0.1 (A linear approximation of a function of two variables)**.** Take $f(\beta) = 3\beta_1 + 5\beta_2^2$ from Example 0.2, whose gradient at $\underset{\sim}{\beta} = (1, 2)^\top$ is $\nabla f(\underset{\sim}{\beta}) = (3, 20)^\top$. At that point,

$$\begin{aligned} f(\underset{\sim}{\beta}) &= 3 \cdot 1 + 5 \cdot 2^2 && \text{(substitute } \beta_1 = 1 \text{ and } \beta_2 = 2) \\ &= 23, && (3 + 20 = 23) \end{aligned}$$

so the linear approximation at $\underset{\sim}{\beta}$ is

$$\hat{f}_{\underset{\sim}{\beta}}(\underset{\sim}{\delta}) = 23 + 3\delta_1 + 20\delta_2.$$

For the step $\underset{\sim}{\delta} = (0.1, 0.1)^\top$, the approximation is

$$\begin{aligned} \hat{f}_{\underset{\sim}{\beta}}(\underset{\sim}{\delta}) &= 23 + 3 \cdot 0.1 + 20 \cdot 0.1 && \text{(substitute } \delta_1 = \delta_2 = 0.1) \\ &= 25.3, && (23 + 0.3 + 2 = 25.3) \end{aligned}$$

and the value of f at $\underset{\sim}{\beta} + \underset{\sim}{\delta} = (1.1, 2.1)^\top$ is

$$\begin{aligned} f(\underset{\sim}{\beta} + \underset{\sim}{\delta}) &= 3 \cdot 1.1 + 5 \cdot 2.1^2 && \text{(substitute } \beta_1 = 1.1 \text{ and } \beta_2 = 2.1) \\ &= 25.35. && (3.3 + 22.05 = 25.35) \end{aligned}$$

The approximation misses by 0.05, which is $5\delta_2^2$, the second-order term of f that the approximation leaves out. That term is never negative, and is zero only when $\delta_2 = 0$, so the graph of the approximation in Figure 1 is a plane that touches the graph of f along the line $\beta_2 = 2$ and lies below it everywhere else.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 1: The graph of $f(\underset{\sim}{\beta}) = 3\beta_1 + 5\beta_2^2$ (blue) and its linear approximation at $(1, 2)^\top$ (vermilion), drawn as a function of the point $(1, 2)^\top + \underset{\sim}{\delta}$. The two surfaces touch along the line $\beta_2 = 2$.

i Remark (The one-variable linear approximation is the case $p = 1$)

Remark 0.1 (The one-variable linear approximation is the case $p = 1$). For $p = 1$, the gradient of f is its derivative, so Equation 1 is the one-variable linear approximation, with the step $\underset{\sim}{\delta}$ in place of ε.

Definition 0.2 (Direction)**.** A **direction** in $\mathbb{R}^p$ is given by a nonzero vector $\underset{\sim}{d} \in \mathbb{R}^p$: moving from a point $\underset{\sim}{x}$ in the direction $\underset{\sim}{d}$ means moving to the points $\underset{\sim}{x} + t\underset{\sim}{d}$ with $t > 0$. For any $c > 0$, $c\underset{\sim}{d}$ reaches the same points (with t/c in place of t), so $\underset{\sim}{d}$ and $c\underset{\sim}{d}$ give the same direction.

Example 0.2 (Moving from $(2, -1)^\top$ in the direction $(1, 1)^\top$)**.** From $\underset{\sim}{x} = (2, -1)^\top$, moving in the direction $\underset{\sim}{d} = (1, 1)^\top$ reaches the points $\underset{\sim}{x} + t\underset{\sim}{d} = (2 + t, -1 + t)^\top$ for

$t > 0$, such as $(3,0)^\top$ at $t = 1$. The vector $(2,2)^\top$ gives the same direction: it reaches $(3,0)^\top$ at $t = \frac{1}{2}$. The vector $(-1,-1)^\top$ gives the opposite direction, reaching points such as $(1,-2)^\top$.

Definition 0.3 (Level set)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$, and let c be a real number. The **level set** [7] of f at c is the preimage of the single value c under f:

$$f^{-1}(\{c\}) \stackrel{\text{def}}{=} f^{-1}(\{c\}).$$

Theorem 0.1 (A level set is the set of points where the function takes the value)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$, and let c be a real number. The level set of f at c is the set of points where f takes the value c:*

$$f^{-1}(\{c\}) = \{\underset{\sim}{x} \in \mathbb{R}^p : f(\underset{\sim}{x}) = c\}.$$

Proof

Proof. Unfold the preimage:

$$\begin{aligned} f^{-1}(\{c\}) &= f^{-1}(\{c\}) && \text{(Definition 0.3)} \\ &= \{\underset{\sim}{x} \in \mathbb{R}^p : f(\underset{\sim}{x}) \in \{c\}\} && \text{(definition of the preimage)} \\ &= \{\underset{\sim}{x} \in \mathbb{R}^p : f(\underset{\sim}{x}) = c\}. && (\{c\} \text{ has the one element } c) \end{aligned}$$

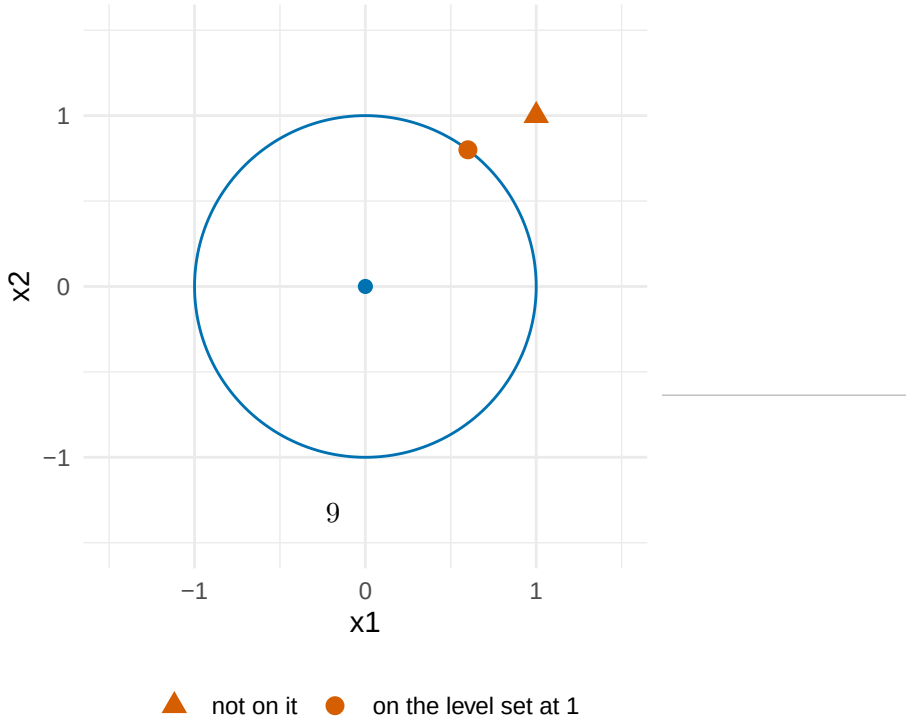
□

Example 0.3 (Level sets of $x_1^2 + x_2^2$)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ for $\underset{\sim}{x} = (x_1, x_2)^\top$. By Theorem 0.1, each level set of f holds the points where f takes that value:

- The level set $f^{-1}(\{1\})$ is the circle of radius 1 around $\underset{\sim}{0}$. For example, $(0.6, 0.8)^\top$ is on it, since $0.36 + 0.64 = 1$, and $(1,1)^\top$ is not, since $\underset{\sim}{1} + 1 = 2$.
- The level set $f^{-1}(\{0\})$ is the single point $\underset{\sim}{0}$.
- The level set $f^{-1}(\{-1\})$ is empty, since $x_1^2 + x_2^2 \geq 0$.

Figure 2 draws the first two.

```r
ls_angle_ls <- seq(0, 2 * pi, length.out = 201)
ls_x1_ls <- cos(ls_angle_ls)
ls_x2_ls <- sin(ls_angle_ls)
ls_circle_ls <- tibble::tibble(x1 = ls_x1_ls, x2 = ls_x2_ls)
ls_origin_ls <- tibble::tibble(x1 = 0, x2 = 0)
ls_kinds_ls <- c("on the level set at 1", "not on it")
ls_points_x1_ls <- c(0.6, 1)
ls_points_x2_ls <- c(0.8, 1)
ls_points_ls <- tibble::tibble(
  x1 = ls_points_x1_ls,
  x2 = ls_points_x2_ls,
  kind = ls_kinds_ls
)
ls_points_aes_ls <- ggplot2::aes(shape = kind)
ls_shapes_ls <- c("on the level set at 1" = 16, "not on it" = 17)
ls_limits_ls <- c(-1.5, 1.5)
ls_circle_ls |>
  ggplot2::ggplot() +
  ggplot2::aes(x = x1, y = x2) +
  ggplot2::geom_path(colour = "#0072B2") +
  ggplot2::geom_point(data = ls_origin_ls, colour = "#0072B2", size = 2) +
  ggplot2::geom_point(
    data = ls_points_ls, mapping = ls_points_aes_ls,
    colour = "#D55E00", size = 3
  ) +
  ggplot2::scale_shape_manual(values = ls_shapes_ls) +
  ggplot2::coord_equal(xlim = ls_limits_ls, ylim = ls_limits_ls) +
  ggplot2::labs(shape = NULL) +
  ggplot2::theme_minimal() +
  ggplot2::theme(legend.position = "bottom")
```



Remark (Level sets in the plane)

Remark 0.2 (Level sets in the plane). For $p = 2$, a level set is often a curve in the plane, such as the circle $f^{-1}(\{1\})$ in Example 0.3.

Definition 0.4 (Level curve)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ and let c be a real number. The **level curve** of f at c is the level set of f at c when it is a curve in the plane. It is the special case $p = 2$ of a level set. A contour line is a level curve drawn on a plot of f.

Example 0.4 (A level curve of $x_1^2 + x_2^2$)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ for $\underset{\sim}{x} = (x_1, x_2)^\top$. The level curve at 4 is the circle of radius 2 around $\underset{\sim}{0}$.

Theorem 0.2 (Level sets of a linear function)**.** *Let $h(\underset{\sim}{x}) = \beta \cdot \underset{\sim}{x} + \beta_0$ with $\beta \neq 0$. Every level set $h^{-1}(\{c\})$ is a hyperplane with normal vector $\underset{\sim}{\beta}$, parallel to the level set $h^{-1}(\{0\})$. Its dimension is $p - 1$: a point for $p = 1$, a line for $p = 2$, and a two-dimensional plane for $p = 3$.*

Proof

Proof. For any c,

$$\begin{aligned} h^{-1}(\{c\}) &= \{\underset{\sim}{x} : \underset{\sim}{\beta} \cdot \underset{\sim}{x} + \beta_0 = c\} \\ &= \{\underset{\sim}{x} : \underset{\sim}{\beta} \cdot \underset{\sim}{x} + (\beta_0 - c) = 0\}. \end{aligned}$$

This is the hyperplane with normal vector $\underset{\sim}{\beta}$ and offset $\beta_0 - c$. Changing c changes only the offset, so the level sets share the normal vector and are parallel. A hyperplane in $\mathbb{R}^p$ has dimension $p - 1$ by the hyperplane theorem, which gives the point, line and plane cases for $p = 1, 2, 3$. □

Example 0.5 (Parallel level lines)**.** Let $h(\underset{\sim}{x}) = 2x_1 + x_2 + 1$, so $\underset{\sim}{\beta} = (2, 1)^\top$ and $\beta_0 = 1$. The level set at c is the line $x_2 = c - 1 - 2x_1$. For $c = -1, 0, 1$ the lines are $x_2 = -2 - 2x_1$, $x_2 = -1 - 2x_1$ and $x_2 = -2x_1$. They all have slope -2 and differ only in the intercept, so they are parallel.

Theorem 0.3 (Stepping along the normal vector)**.** *Let $h(\underset{\sim}{x}) = \underset{\sim}{\beta} \cdot \underset{\sim}{x} + \beta_0$ with $\underset{\sim}{\beta} \neq \underset{\sim}{0}$. For every $\underset{\sim}{x} \in \mathbb{R}^p$ and $t \in \mathbb{R}$,*

$$h(\underset{\sim}{x} + t\underset{\sim}{\beta}) = h(\underset{\sim}{x}) + t\|\underset{\sim}{\beta}\|^2.$$

So a step along $\underset{\sim}{\beta}$ with $t > 0$ increases h.

Proof

Proof. By linearity of the dot product,

$$\begin{aligned} h(\underset{\sim}{x} + t\underset{\sim}{\beta}) &= \underset{\sim}{\beta} \cdot \underset{\sim}{x} + t\underset{\sim}{\beta} + \beta_0 \\ &= \underset{\sim}{\beta} \cdot \underset{\sim}{x} + t\underset{\sim}{\beta} \cdot \underset{\sim}{\beta} + \beta_0 \\ &= h(\underset{\sim}{x}) + t\|\underset{\sim}{\beta}\|^2. \end{aligned}$$

□

Example 0.6 (A step along the normal vector)**.** Let $h(\underset{\sim}{x}) = 2x_1 + x_2 + 1$, so $\beta = (2, 1)^\top$ and $\|\beta\|^2 = 5$. At $\underset{\sim}{x} = \underset{\sim}{0}$, $h(\underset{\sim}{0}) = 1$. A step of $t = 1$ along $\underset{\sim}{\beta}$ lands at $(2, \widetilde{1})^\top$, where $h\big((2, \widetilde{1})^\top\big) = 4 + \widetilde{1} + 1 = \widetilde{6} = \widetilde{1} + 5$.

Theorem 0.4 (The normal vector is the steepest direction)**.** *Let $h(\underset{\sim}{x}) = \beta \cdot \underset{\sim}{x} + \beta_0$ with $\beta \neq \underset{\sim}{0}$. Among unit vectors $\underset{\sim}{u}$ ($\|\underset{\sim}{u}\| = 1$), the change $h(\underset{\sim}{x} + t\underset{\sim}{u}) - h(\underset{\sim}{x})$ with $\widetilde{t} > 0$ is largest for $\underset{\sim}{u} = \underset{\sim}{\beta}/\|\underset{\sim}{\beta}\|$.*

Proof

Proof. For a unit vector $\underset{\sim}{u}$,

$$h(\underset{\sim}{x} + t\underset{\sim}{u}) - h(\underset{\sim}{x}) = t\underset{\sim}{\beta} \cdot \underset{\sim}{u}.$$

Since $t > 0$, maximizing this change is the same as maximizing $\beta \cdot \underset{\sim}{u}$. By the Cauchy-Schwarz inequality, $\beta \cdot \underset{\sim}{u} \leq \|\beta\|\|\underset{\sim}{u}\| = \|\beta\|$. Equality holds exactly when $\underset{\sim}{u} = \alpha\underset{\sim}{\beta}$ for some $\alpha > 0$. The condition $\|\underset{\sim}{u}\| = \widetilde{1}$ then forces $\alpha\|\underset{\sim}{\beta}\| = 1$, so

$$\underset{\sim}{u} = \underset{\sim}{\beta}/\|\underset{\sim}{\beta}\|.$$

□

Example 0.7 (Two unit steps compared)**.** Let $h(\underset{\sim}{x}) = 2x_1 + x_2 + 1$, so $\beta = (2, 1)^\top$ and $\|\beta\| = \sqrt{5}$. A step of size 1 along the unit vector $\beta/\sqrt{5}$ raises h by $\underset{\sim}{\beta} \cdot \underset{\sim}{\beta}/\sqrt{5} = \sqrt{5} \approx 2.24$. A step of size 1 along $(1, 0)^\top$ raises it by only $\underset{\sim}{\beta} \cdot (1, 0)^\top = 2$.

The course website's version of this page has an interactive plot here.

The course website's version of this page has an interactive 3D plot here.

Exercise 0.1 (Take a gradient by hand)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be

$$f(\underset{\sim}{w}) = 3w_1w_2 + w_2^3 - 5w_1$$

1. Compute $\partial f/\partial w_1$ (Definition 0.1), treating w_2 as a constant.
2. Compute $\partial f/\partial w_2$, treating w_1 as a constant.
3. Assemble $\nabla_{\underset{\sim}{w}} f$ and evaluate it at $\underset{\sim}{w} = \begin{bmatrix} 2 & -1 \end{bmatrix}^\top$.
4. Check the shape of your answer against Table 1.

Solution

Solution 0.1. **1.** Differentiate with respect to w_1, holding w_2 fixed. The first term is the constant $3w_2$ times w_1, so it contributes $3w_2$. The second term has no w_1 in it, so it contributes 0. The third term is -5 times w_1, so it contributes -5:

$$\begin{aligned} \frac{\partial f}{\partial w_1} &= 3w_2 + 0 - 5 \\ &= 3w_2 - 5 \end{aligned}$$

2. Now with respect to w_2, holding w_1 fixed. The first term is the constant $3w_1$ times w_2, contributing $3w_1$. The second term contributes $3w_2^2$. The third has no w_2 in it, contributing 0:

$$\begin{aligned} \frac{\partial f}{\partial w_2} &= 3w_1 + 3w_2^2 + 0 \\ &= 3w_1 + 3w_2^2 \end{aligned}$$

3. Stack the two, in the order the coordinates are numbered:

$$\nabla_{\underset{\sim}{w}} f(\underset{\sim}{w}) = \begin{bmatrix} 3w_2 - 5 \\ 3w_1 + 3w_2^2 \end{bmatrix}$$

The gradient is a vector-valued *function* of $\underset{\sim}{w}$, not a single vector. At $\underset{\sim}{w} = \begin{bmatrix} 2 & -1 \end{bmatrix}^\top$,

$$\begin{aligned} \nabla_{\underset{\sim}{w}} f &= \begin{bmatrix} 3(-1) - 5 \\ 3(2) + 3(-1)^2 \end{bmatrix} \\ &= \begin{bmatrix} -3 - 5 \\ 6 + 3 \end{bmatrix} \\ &= \begin{bmatrix} -8 \\ 9 \end{bmatrix} \end{aligned}$$

4. The input was a vector in $\mathbb{R}^2$ and so is the answer, as Table 1 requires. Note the signs: from this point, increasing w_1 *decreases* f while increasing w_2 increases it, so the uphill direction is neither axis.

Definition 0.5 (Row-vector derivative)**.** If $f(\underset{\sim}{\beta})$ is a scalar-valued function of a $p \times 1$ vector β, such as $f(\underset{\sim}{\beta}) = \underset{\sim}{x} \cdot \underset{\sim}{\beta}$, then its **row-vector derivative** is the row vector of its partial derivatives (Definition 0.1):

$$\frac{\partial f(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}^{\top}} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\underset{\sim}{\beta}) & \frac{\partial}{\partial \beta_2} f(\underset{\sim}{\beta}) & \cdots & \frac{\partial}{\partial \beta_p} f(\underset{\sim}{\beta}) \end{bmatrix} \tag{2}$$

When the derivative of a scalar-valued function is written as a row vector, it is also called the total derivative [8] of f.

Remark (Other notation for the row-vector derivative)

Remark 0.3 (Other notation for the row-vector derivative). Some sources write the same row vector as $\frac{\partial}{\partial \underset{\sim}{\beta}^{\top}} f(\underset{\sim}{\beta})$, with the operator on the left, read as each operator $\frac{\partial}{\partial \beta_i}$ applied to f rather than as a matrix product (see Remark 0.4).

Remark (Which side the operator goes on)

Remark 0.4 (Which side the operator goes on). Read $\frac{\partial}{\partial \underset{\sim}{\beta}}$ as a $p \times 1$ column vector of operators with entries $\frac{\partial}{\partial \beta_1}, \ldots, \frac{\partial}{\partial \beta_p}$, and $\frac{\partial}{\partial \underset{\sim}{\beta}^{\top}}$ as the $1 \times p$ row vector of operators with the same entries. Writing either one next to f can be read in two ways:

- **As a matrix product,** with f a 1×1 matrix, under the shape rule of matrix multiplication. The column-vector operator works on the left: $\frac{\partial}{\partial \underset{\sim}{\beta}} f$ is $(p \times 1)(1 \times 1)$, a $p \times 1$ column vector (Definition 0.2). The row-vector operator does not: $\frac{\partial}{\partial \underset{\sim}{\beta}^{\top}} f$ is $(1 \times p)(1 \times 1)$, whose inner dimensions p and 1 do not match when $p > 1$. For the shape rule to give the $1 \times p$ row vector, the operator would have to go on the right of f, as in $(1 \times 1)(1 \times p)$, which changes its meaning (see below).
- **As a scalar multiple,** with f a scalar, the way $c\underset{\sim}{x}$ is read in scalar multiplication, which has no shape rule: each entry of the operator vector is applied to f. Then both $\frac{\partial}{\partial \underset{\sim}{\beta}} f$ and $\frac{\partial}{\partial \underset{\sim}{\beta}^{\top}} f$ work with the operator on the left.

The fraction $\frac{\partial f}{\partial \underset{\sim}{\beta}^{\top}}$ of Equation 2 means the same thing under both readings: the transpose in the denominator says the result is a row vector, without making the derivative a product.

Treating the operators as vectors is sound for checking shapes, but "multiplying" an operator by a function means applying the operator to it, so two rules of ordinary matrix algebra do not carry over: scalars commute with matrices, and products can be regrouped.

- **Scalars do not commute with operators.** $\frac{\partial}{\partial \beta_i} f$ is a function, but $f \frac{\partial}{\partial \beta_i}$ is still an unapplied operator, waiting for something to act on. So, unlike $c\underset{\sim}{x} = \underset{\sim}{x}c$, the

operator cannot move to the other side of f without changing the meaning. Putting the row-vector operator on the right of f gives the row-vector derivative only under the convention that it acts on the factor to its left.

- **Products cannot be regrouped.** An operator acts on the whole product to its right, so by the product rule, $\frac{\partial}{\partial \beta_i}(fg) \neq \left(\frac{\partial}{\partial \beta_i} f\right) g$ in general. For example, with $f(\underset{\sim}{\beta}) = \beta_1$, $\frac{\partial}{\partial \beta_1}(ff) = 2\beta_1$ but $\left(\frac{\partial}{\partial \beta_1} f\right) f = \beta_1$.

Theorem 0.5 (Row and column derivatives are transposes)**.**

$$\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}^\top} = \left(\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}}\right)^\top \tag{3}$$

$$\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}} = \left(\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}^\top}\right)^\top \tag{4}$$

Proof

Proof. By Definition 0.2 and Definition 0.5, entry j of both $\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}}$ and $\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}^\top}$ is $\frac{\partial}{\partial \beta_j} f(\underset{\sim}{\beta})$; the first is a $p \times 1$ column vector and the second a $1 \times p$ row vector with the same entries in the same order, so each is the transpose of the other. □

Example 0.8 (Row and column derivatives of a linear function)**.** For $f(\underset{\sim}{\beta}) = 3\beta_1 + 5\beta_2$:

$$\frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \qquad \frac{\partial f(\beta)}{\partial \underset{\sim}{\beta}^\top} = \begin{bmatrix} 3 & 5 \end{bmatrix},$$

and each is the transpose of the other.

Definition 0.6 (Derivative of a vector-valued function)**.** If

$$\begin{aligned} \underset{\sim}{y} &= \underset{\sim}{y}(\underset{\sim}{\beta}) \\ &= (y_1, \ldots, y_q)^\top \end{aligned}$$

is a $q \times 1$ vector-valued function of the $p \times 1$ vector $\underset{\sim}{\beta}$, its **derivative with respect to** $\underset{\sim}{\beta}$ is the $p \times q$ matrix whose (i, j) entry is

$$\left[\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{y}^\top\right]_{ij} \stackrel{\text{def}}{=} \frac{\partial}{\partial \beta_i} y_j, \qquad i = 1, \ldots, p, \quad j = 1, \ldots, q.$$

Both $\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{y}$ and $\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{y}^\top$ denote this $p \times q$ matrix.

Definition 0.7 (Denominator layout)**.** Writing derivatives as in Definition 0.6 is the **denominator layout** [9], which these notes use throughout: rows index the entries of β (the denominator) and columns index the entries of $\underset{\sim}{y}$ (the numerator), so column j is the vector derivative $\frac{\partial}{\partial \underset{\sim}{\beta}} y_j$ (Definition 0.2).

Example 0.9 (Differentiating a 3×1 function of a 2×1 vector)**.** Let $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$ ($p = 2$) and $\underset{\sim}{y}(\underset{\sim}{\beta}) = (\beta_1^2,\ \beta_1\beta_2,\ 3\beta_2)^\top$ ($q = 3$). Then

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{y}^\top}_{2\times 3} = \begin{bmatrix} \frac{\partial}{\partial \beta_1}\beta_1^2 & \frac{\partial}{\partial \beta_1}\beta_1\beta_2 & \frac{\partial}{\partial \beta_1}3\beta_2 \\ \frac{\partial}{\partial \beta_2}\beta_1^2 & \frac{\partial}{\partial \beta_2}\beta_1\beta_2 & \frac{\partial}{\partial \beta_2}3\beta_2 \end{bmatrix} = \begin{bmatrix} 2\beta_1 & \beta_2 & 0 \\ 0 & \beta_1 & 3 \end{bmatrix}$$

Definition 0.8 (Jacobian matrix)**.** Let

$$\underset{\sim}{y} = \underset{\sim}{y}(\underset{\sim}{\beta}) = (y_1, \dots, y_q)^\top$$

be a $q \times 1$ vector-valued function of the $p \times 1$ vector β. The **Jacobian matrix** [10] of y is the $q \times p$ matrix whose (j, i) entry is $\frac{\partial}{\partial \beta_i} y_j$, for $j = 1, \dots, q$ and $i = 1, \dots, p$. It is the transpose of the derivative $\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{y}^\top$ of Definition 0.6.

Definition 0.9 (Numerator layout)**.** Writing derivatives as in Definition 0.8, with rows indexing the entries of y (the numerator) and columns indexing the entries of $\underset{\sim}{\beta}$ (the denominator), is the **numerator layout** [9].

Example 0.10 (The Jacobian matrix of a 3×1 function of a 2×1 vector)**.** For $\underset{\sim}{y}(\underset{\sim}{\beta}) = (\beta_1^2,\ \beta_1\beta_2,\ 3\beta_2)^\top$ of Example 0.9, the Jacobian matrix is the 3×2 matrix

$$\begin{bmatrix} \frac{\partial}{\partial \beta_1}\beta_1^2 & \frac{\partial}{\partial \beta_2}\beta_1^2 \\ \frac{\partial}{\partial \beta_1}\beta_1\beta_2 & \frac{\partial}{\partial \beta_2}\beta_1\beta_2 \\ \frac{\partial}{\partial \beta_1}3\beta_2 & \frac{\partial}{\partial \beta_2}3\beta_2 \end{bmatrix} = \begin{bmatrix} 2\beta_1 & 0 \\ \beta_2 & \beta_1 \\ 0 & 3 \end{bmatrix},$$

the transpose of the 2×3 matrix found there.

Remark (Numerator layout)

Remark 0.5 (Numerator layout). Some sources use the numerator layout (Definition 0.9) throughout, so that their derivative of a vector-valued function is the Jacobian matrix, the transpose of the derivative used in these notes. In numerator layout, the derivative of a scalar-valued function is a row vector: the row-vector derivative (Definition 0.5). Check a source's layout before combining its formulas with these.

Definition 0.10 (Constant)**.** A $q \times 1$ vector $\underset{\sim}{x}$ is **constant with respect to** the $p \times 1$ vector $\underset{\sim}{\beta}$ if its derivative (Definition 0.6) is zero:

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{x}^\top}_{p \times q} = \underset{p \times q}{\mathbf{0}}$$

Example 0.11 (A constant vector)**.** Let $\beta = (\beta_1, \beta_2)^\top$ and $\underset{\sim}{x} = (3, 5)^\top$, so $x_1 = 3$ and $x_2 = 5$ do not depend on β. Expanding $\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{x}^\top$ into its matrix of scalar partial derivatives (Definition 0.6) and evaluating each entry:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{x}^\top}_{2 \times 2} &= \frac{\partial}{\partial \underset{\sim}{\beta}} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \\
&= \begin{bmatrix} \frac{\partial}{\partial \beta_1} x_1 & \frac{\partial}{\partial \beta_1} x_2 \\ \frac{\partial}{\partial \beta_2} x_1 & \frac{\partial}{\partial \beta_2} x_2 \end{bmatrix} \\
&= \begin{bmatrix} \frac{\partial}{\partial \beta_1} 3 & \frac{\partial}{\partial \beta_1} 5 \\ \frac{\partial}{\partial \beta_2} 3 & \frac{\partial}{\partial \beta_2} 5 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\
&= \underset{2 \times 2}{\mathbf{0}}
\end{aligned}$$

Every entry is the derivative of a constant, so $\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{x}^\top = \underset{2 \times 2}{\mathbf{0}}$ and $\underset{\sim}{x}$ is constant with respect to $\underset{\sim}{\beta}$ (Definition 0.10).

Example 0.12 (A vector that is not constant)**.** With the same $\underset{\sim}{\beta}$, let $\underset{\sim}{x} = (\beta_1, 3)^\top$. Then

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{x}^\top}_{2\times 2} = \begin{bmatrix} \frac{\partial}{\partial \beta_1}\beta_1 & \frac{\partial}{\partial \beta_1}3 \\ \frac{\partial}{\partial \beta_2}\beta_1 & \frac{\partial}{\partial \beta_2}3 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

which is not $\underset{2\times 2}{\underset{\sim}{\mathbf{0}}}$, so this $\underset{\sim}{x}$ is not constant with respect to $\underset{\sim}{\beta}$: its first entry changes when β_1 does.

Exercise 0.2 (Partial derivative with respect to an unreferenced variable)**.** Adapted from Miller [3], Question 1.1.46.
Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for

$$f(x, y, t) = 5t^4 - 4t^5 \cos(t \sin t)$$

Solution

Solution 0.2. Although the function is declared as a function of three variables (x, y, t), its expression depends solely on t.
Because neither x nor y appears in the formula, f is constant with respect to both x and y:

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 0$$

Remark: Always verify which variables actually appear in a formula before performing lengthy algebraic differentiation.

Theorem 0.6 (Derivative of a dot product)**.** *If $\underset{\sim}{x}$ is constant with respect to $\underset{\sim}{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{x} \cdot \underset{\sim}{\beta})}_{p\times 1} = \underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta} \cdot \underset{\sim}{x})}_{p\times 1}$$
$$= \underset{p\times 1}{\underset{\sim}{x}}$$

Proof

Proof.

$$\begin{aligned}\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{x}\cdot\underset{\sim}{\beta}) &= \begin{bmatrix}\frac{\partial}{\partial\beta_1}(x_1\beta_1+x_2\beta_2+\ldots+x_p\beta_p)\\ \frac{\partial}{\partial\beta_2}(x_1\beta_1+x_2\beta_2+\ldots+x_p\beta_p)\\ \vdots \\ \frac{\partial}{\partial\beta_p}(x_1\beta_1+x_2\beta_2+\ldots+x_p\beta_p)\end{bmatrix}\\ &= \begin{bmatrix}x_1\\x_2\\\vdots\\x_p\end{bmatrix}\\ &= \underset{\sim}{x}\end{aligned}$$

□

Example 0.13 (Derivative of a dot product)**.** Let $\underset{\sim}{x} = (3,5)^\top$ (constant with respect to $\underset{\sim}{\beta}$; see Example 0.11) and $\underset{\sim}{\beta} = (\beta_1,\beta_2)^\top$. Then $\underset{\sim}{x}\cdot\underset{\sim}{\beta} = 3\beta_1+5\beta_2$, and by Theorem 0.6:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial\underset{\sim}{\beta}}(\underset{\sim}{x}\cdot\underset{\sim}{\beta})}_{2\times1} &= \underset{2\times1}{\underset{\sim}{x}}\\ &= \begin{pmatrix}3\\5\end{pmatrix}\end{aligned}$$

Verifying entry-wise:

$$\begin{aligned}\frac{\partial}{\partial\underset{\sim}{\beta}}(3\beta_1+5\beta_2) &= \begin{pmatrix}\frac{\partial}{\partial\beta_1}(3\beta_1+5\beta_2)\\ \frac{\partial}{\partial\beta_2}(3\beta_1+5\beta_2)\end{pmatrix}\\ &= \begin{pmatrix}3\\5\end{pmatrix}\end{aligned}$$

Both methods agree.

Theorem 0.7 (Product rule for dot-products)**.** *If $\underset{\sim}{a} = \underset{\sim}{a}(\underset{\sim}{x})$ and $\underset{\sim}{b} = \underset{\sim}{b}(\underset{\sim}{x})$ are differentiable $p\times1$ vector functions of $\underset{\sim}{x}$, then:*

$$\frac{\partial}{\partial\underset{p\times1}{\underset{\sim}{x}}}\underset{p\times1}{\underset{\sim}{a}}\cdot\underset{p\times1}{\underset{\sim}{b}} = \left(\frac{\partial}{\partial\underset{p\times1}{\underset{\sim}{x}}}\underset{1\times p}{\underset{\sim}{a}^\top}\right)\underset{p\times1}{\underset{\sim}{b}} + \left(\frac{\partial}{\partial\underset{p\times1}{\underset{\sim}{x}}}\underset{1\times p}{\underset{\sim}{b}^\top}\right)\underset{p\times1}{\underset{\sim}{a}}$$

Proof

Proof. Entry-wise, for $i = 1, \dots, p$:

$$\begin{aligned}\left[\frac{\partial}{\partial \underset{\sim}{x}}(\underset{\sim}{a} \cdot \underset{\sim}{b})\right]_i &= \frac{\partial}{\partial x_i}\sum_{k=1}^{p} a_k b_k \\ &= \sum_{k=1}^{p}\left(b_k \frac{\partial}{\partial x_i} a_k + a_k \frac{\partial}{\partial x_i} b_k\right) \\ &= \left[\left(\frac{\partial}{\partial \underset{\sim}{x}} \underset{\sim}{a}^\top\right)\underset{\sim}{b}\right]_i + \left[\left(\frac{\partial}{\partial \underset{\sim}{x}} \underset{\sim}{b}^\top\right)\underset{\sim}{a}\right]_i\end{aligned}$$

□

Example 0.14 (Example of the dot-product rule)**.** Apply Theorem 0.7 with the vector $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$ in the role of $\underset{\sim}{x}$. Let $\underset{\sim}{a}(\underset{\sim}{\beta}) = (\beta_1, \beta_1\beta_2)^\top$ and $\underset{\sim}{b}(\underset{\sim}{\beta}) = (\beta_2, \beta_1)^\top$. Then:

$$\begin{aligned}\underset{\sim}{a} \cdot \underset{\sim}{b} &= \beta_1 \cdot \beta_2 + \beta_1\beta_2 \cdot \beta_1 \\ &= \beta_1\beta_2 + \beta_1^2\beta_2\end{aligned}$$

By direct calculation:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{a} \cdot \underset{\sim}{b})}_{2\times 1} &= \frac{\partial}{\partial \underset{\sim}{\beta}}(\beta_1\beta_2 + \beta_1^2\beta_2) \\ &= \begin{pmatrix}\beta_2 + 2\beta_1\beta_2 \\ \beta_1 + \beta_1^2\end{pmatrix}\end{aligned}$$

By the product rule (Theorem 0.7), using $\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{a}^\top}_{2\times 2} = \begin{pmatrix}1 & \beta_2 \\ 0 & \beta_1\end{pmatrix}$ and $\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}} \underset{\sim}{b}^\top}_{2\times 2} = \begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix}$:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{a} \cdot \underset{\sim}{b})}_{2\times 1} &= \underbrace{\begin{pmatrix}1 & \beta_2 \\ 0 & \beta_1\end{pmatrix}}_{2\times 2}\underbrace{\begin{pmatrix}\beta_2 \\ \beta_1\end{pmatrix}}_{2\times 1} + \underbrace{\begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix}}_{2\times 2}\underbrace{\begin{pmatrix}\beta_1 \\ \beta_1\beta_2\end{pmatrix}}_{2\times 1} \\ &= \begin{pmatrix}\beta_2 + \beta_1\beta_2 \\ \beta_1^2\end{pmatrix} + \begin{pmatrix}\beta_1\beta_2 \\ \beta_1\end{pmatrix} \\ &= \begin{pmatrix}\beta_2 + 2\beta_1\beta_2 \\ \beta_1^2 + \beta_1\end{pmatrix}\end{aligned}$$

Both methods agree.

Theorem 0.8 (Derivative of a linear map)**.** *If* $\mathbf{A}$ *is an* $m \times p$ *matrix that is constant with respect to* $\underset{\sim}{\beta}$*, then:*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{\beta})}_{p\times m} = \underset{p\times m}{\underset{\sim}{\mathbf{A}^\top}}$$

Proof

Proof. For entry (i, j), where row i indexes the denominator $\underset{\sim}{\beta}$ (see Definition 0.6) and column j indexes the numerator $\mathbf{A}\underset{\sim}{\beta}$:

$$\begin{aligned}\left[\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{\beta})\right]_{ij} &= \frac{\partial}{\partial \beta_i}(\mathbf{A}\underset{\sim}{\beta})_j \\ &= \frac{\partial}{\partial \beta_i}\sum_{k=1}^{p} a_{jk}\beta_k \\ &= a_{ji} \\ &= [\mathbf{A}^\top]_{ij}\end{aligned}$$

□

Example 0.15 (Derivative of a linear map)**.** Let $\mathbf{A} = \begin{pmatrix}2 & 3\end{pmatrix}$ (1×2) and $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{A}\underset{\sim}{\beta} = 2\beta_1 + 3\beta_2$, and by Theorem 0.8:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{\beta})}_{2\times 1} &= \underset{2\times 1}{\underset{\sim}{\mathbf{A}^\top}} \\ &= \begin{pmatrix}2\\3\end{pmatrix}\end{aligned}$$

Theorem 0.9 (Vector-derivative of a matrix-vector product)**.** *If* $\mathbf{A}$ *is an* $m \times q$ *matrix that is constant with respect to* $\underset{\sim}{\beta}$*, and* $\underset{\sim}{v} = \underset{\sim}{v}(\underset{\sim}{\beta})$ *is a* $q \times 1$ *vector that depends on the* $p \times 1$ *vector* $\underset{\sim}{\beta}$*, then:*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{v})}_{p\times m} = \underbrace{\left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{v}\right)}_{p\times q}\underset{q\times m}{\underset{\sim}{\mathbf{A}^\top}}$$

Proof

Proof. For entry (i, j), where row i indexes the denominator $\underset{\sim}{\beta}$ and column j indexes the numerator $\mathbf{A}\underset{\sim}{v}$ (see Definition 0.6):

$$\begin{aligned}\left[\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{v})\right]_{ij} &= \frac{\partial}{\partial \beta_i}(\mathbf{A}\underset{\sim}{v})_j \\ &= \frac{\partial}{\partial \beta_i}\sum_{k=1}^{q} a_{jk}v_k \\ &= \sum_{k=1}^{q} a_{jk}\frac{\partial}{\partial \beta_i}v_k \\ &= \sum_{k=1}^{q}\left[\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{v}\right]_{ik}[\mathbf{A}^\top]_{kj} \\ &= \left[\left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{v}\right)\mathbf{A}^\top\right]_{ij}\end{aligned}$$

□

Example 0.16 (Vector-derivative of a matrix-vector product)**.** Let $\mathbf{A} = \begin{pmatrix}2 & 3\end{pmatrix}$ (1×2, constant) and $\underset{\sim}{v}(\underset{\sim}{\beta}) = (\beta_1^2, \beta_2^2)^\top$. Then $\mathbf{A}\underset{\sim}{v} = 2\beta_1^2 + 3\beta_2^2$. By Theorem 0.9:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{v})}_{2\times 1} &= \begin{pmatrix}2\beta_1 & 0 \\ 0 & 2\beta_2\end{pmatrix}\begin{pmatrix}2 \\ 3\end{pmatrix} \\ &= \begin{pmatrix}4\beta_1 \\ 6\beta_2\end{pmatrix}\end{aligned}$$

ℹ Remark (The derivative of a linear map as a special case)

Remark 0.6 (The derivative of a linear map as a special case). This result generalizes Theorem 0.8, which is the special case $\underset{\sim}{v} = \underset{\sim}{\beta}$ (so that $q = p$, $\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\beta} = \mathbf{I}_p$, and $\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\underset{\sim}{\beta}) = \mathbf{I}_p\mathbf{A}^\top$, which is $\mathbf{A}^\top$). For example, Example 0.15 is the case $\mathbf{A} = \begin{pmatrix}2 & 3\end{pmatrix}$ and

$$\begin{aligned}\underset{\sim}{v} &= \underset{\sim}{\beta} \\ &= (\beta_1, \beta_2)^\top,\end{aligned}$$

where this result gives $\mathbf{I}_2\mathbf{A}^\top = (2, 3)^\top$.

Theorem 0.10 (Vector-derivative of a product of matrices)**.** *If* $\mathbf{A}$ *($k \times m$) and* $\mathbf{B}$ *($m \times q$) are constant with respect to* $\underset{\sim}{\beta}$*, and* $\underset{\sim}{v} = \underset{\sim}{v}(\underset{\sim}{\beta})$ *is a* $q \times 1$ *vector that depends on the* $p \times 1$ *vector* $\underset{\sim}{\beta}$*, then:*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\mathbf{B}\underset{\sim}{v})}_{p\times k} = \underbrace{\left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{v}\right)}_{p\times q} \underbrace{\mathbf{B}^\top}_{q\times m}\,\underbrace{\mathbf{A}^\top}_{m\times k}$$

Proof

Proof. Apply Theorem 0.9 with the constant $k \times q$ matrix $\mathbf{AB}$, then use the transpose of a product:

$$\begin{aligned}\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\mathbf{B}\underset{\sim}{v}) &= \left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{v}\right)(\mathbf{A}\mathbf{B})^\top \quad \text{(derivative of a matrix-vector product)}\\ &= \left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{v}\right)\mathbf{B}^\top\mathbf{A}^\top \quad \text{(transpose of a product)}\end{aligned}$$

□

Example 0.17. Let $\mathbf{A} = \begin{pmatrix}1 & 0\end{pmatrix}$ (1×2), $\mathbf{B} = \begin{pmatrix}2 & 0\\ 0 & 3\end{pmatrix}$ (2×2), and $\underset{\sim}{v}(\underset{\sim}{\beta}) = \underset{\sim}{\beta}$ where $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{AB}\underset{\sim}{v} = 2\beta_1$, and:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{A}\mathbf{B}\underset{\sim}{v})}_{2\times 1} &= \underbrace{\left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\beta}\right)}_{2\times 2}\underbrace{\mathbf{B}^\top}_{2\times 2}\underbrace{\mathbf{A}^\top}_{2\times 1}\\ &= \mathbf{I}_2\begin{pmatrix}2 & 0\\ 0 & 3\end{pmatrix}\begin{pmatrix}1\\ 0\end{pmatrix}\\ &= \begin{pmatrix}2\\ 0\end{pmatrix}\end{aligned}$$

Corollary 0.1 (Derivative of a dot product, transpose-product form)**.** *If $\underset{\sim}{x}$ is constant with respect to $\underset{\sim}{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underbrace{\underset{\sim}{x}^\top}_{1\times p}\underbrace{\underset{\sim}{\beta}}_{p\times 1})}_{p\times 1} = \underbrace{\underset{\sim}{x}}_{p\times 1}$$

Proof

Proof. **Using Theorem 0.6:**
Since $\underset{\sim}{x}^\top\underset{\sim}{\beta} = \underset{\sim}{x}\cdot\underset{\sim}{\beta}$ (see dot product), and $\underset{\sim}{x}$ is constant with respect to $\underset{\sim}{\beta}$:

$$\begin{aligned}\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{x}^\top \underset{\sim}{\beta}) &= \frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{x} \cdot \underset{\sim}{\beta}) \\ &= \underset{\sim}{x}\end{aligned}$$

by Theorem 0.6. □

Proof

Proof. **Using Theorem 0.9:**
Since $\underset{\sim}{x}$ is constant with respect to β, $\mathbf{A} = \underset{\sim}{x}^\top$ is a constant $1 \times p$ matrix. Applying Theorem 0.9 with $\underset{\sim}{v} = \underset{\sim}{\beta}$ (so $\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\beta} = \mathbf{I}$):

$$\begin{aligned}\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{x}^\top \underset{\sim}{\beta}) &= \left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\beta}\right)(\underset{\sim}{x}^\top)^\top \\ &= \mathbf{I} \cdot \underset{\sim}{x} \\ &= \underset{\sim}{x}\end{aligned}$$

□

Example 0.18 (Derivative of a transpose product)**.** Let $\underset{\sim}{x} = (3, 5)^\top$ and $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$. Then $\underset{\sim}{x}^\top \underset{\sim}{\beta} = 3\beta_1 + 5\beta_2$, and by Corollary 0.1:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}\left(\underset{1\times 2}{\underset{\sim}{x}^\top}\ \underset{2\times 1}{\underset{\sim}{\beta}}\right)}_{2\times 1} &= \underset{2\times 1}{\underset{\sim}{x}} \\ &= \begin{pmatrix}3\\5\end{pmatrix}\end{aligned}$$

Remark (The coefficient gets transposed)

Remark 0.7 (The coefficient gets transposed). This vector derivative formula looks a lot like non-vector calculus, except that you have to transpose the coefficient: in scalar calculus $\frac{\partial}{\partial x}(cx) = c$, but here the coefficient $\underset{\sim}{x}^\top$ (a row vector) becomes $\underset{\sim}{x}$ (a column vector) in the result. For example, with $\underset{\sim}{x} = (2, -1)^\top$, $\underset{\sim}{x}^\top \underset{\sim}{\beta} = 2\beta_1 - \beta_2$, whose vector derivative is the column vector $(2, -1)^\top = \underset{\sim}{x}$, not the row vector $\underset{\sim}{x}^\top = (2, -1)$.

Theorem 0.11 (Derivative of a quadratic form)**.** *For a quadratic form (see quadratic form), if* $\mathbf{S}$ *is a* $p \times p$ *matrix that is constant with respect to* $\underset{\sim}{\beta}$*, then:*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta})}_{p\times 1} = \underbrace{(\mathbf{S}+\mathbf{S}^\top)\underset{\sim}{\beta}}_{p\times 1}.$$

Proof

Proof. Expanding entry-wise, $\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta} = \sum_{j=1}^p \sum_{k=1}^p s_{jk}\beta_j\beta_k$. Differentiating component-wise with respect to β_i for $i = 1, \ldots, p$:

$$\begin{aligned}
\left[\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta})\right]_i &= \frac{\partial}{\partial \beta_i} \sum_{j=1}^p \sum_{k=1}^p s_{jk}\beta_j\beta_k && \text{(expand quadratic form)} \\
&= \sum_{k=1}^p s_{ik}\beta_k + \sum_{j=1}^p s_{ji}\beta_j && \text{(product rule for } \beta_i\beta_k\text{)} \\
&= [\mathbf{S}\underset{\sim}{\beta}]_i + [\mathbf{S}^\top \underset{\sim}{\beta}]_i && \text{(matrix-vector multiplication definition)} \\
&= [(\mathbf{S}+\mathbf{S}^\top)\underset{\sim}{\beta}]_i && \text{(linearity of matrix multiplication)}
\end{aligned}$$

□

Corollary 0.2 (Derivative of a quadratic form with a symmetric matrix)**.** *If, in addition,* $\mathbf{S}$ *is symmetric (and, as in Theorem 0.11,* $p \times p$ *and constant with respect to* $\underset{\sim}{\beta}$*), then*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta})}_{p\times 1} = \underbrace{2\mathbf{S}\underset{\sim}{\beta}}_{p\times 1}.$$

Proof

Proof. If $\mathbf{S} = \mathbf{S}^\top$, then $\mathbf{S} + \mathbf{S}^\top = 2\mathbf{S}$, so Theorem 0.11 gives $\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta}) = 2\mathbf{S}\underset{\sim}{\beta}$. □

Remark (Like the derivative of cx^2)

Remark 0.8 (Like the derivative of cx^2). This operation is like taking the derivative of cx^2 with respect to x in non-vector calculus: $\frac{\partial}{\partial x}(cx^2) = 2cx$, and Corollary 0.2 says $\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta}) = 2\mathbf{S}\underset{\sim}{\beta}$. For example, with $p = 1$, $\mathbf{S} = (3)$, and $\underset{\sim}{\beta} = (\beta_1)$, $\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta} = 3\beta_1^2$, and its derivative is $6\beta_1 = 2\mathbf{S}\underset{\sim}{\beta}$.

Example 0.19 (Derivative of a quadratic form)**.** Let $\mathbf{S} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ (2×2, symmetric and constant) and $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$. Then $\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta} = 3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2$. By Corollary 0.2:

$$\underbrace{\frac{\partial}{\partial \beta}(\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta})}_{2\times 1} = 2\mathbf{S}\underset{\sim}{\beta}$$
$$= 2\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$
$$= \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Differentiating component-wise directly:

$$\begin{pmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Both methods agree.

For a matrix that is not symmetric, such as $\mathbf{S} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$, Theorem 0.11 gives

$$(\mathbf{S} + \mathbf{S}^\top)\underset{\sim}{\beta} = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$
$$= \begin{pmatrix} 2\beta_1 + 2\beta_2 \\ 2\beta_1 + 6\beta_2 \end{pmatrix},$$

which matches differentiating $\underset{\sim}{\beta}^\top \mathbf{S} \underset{\sim}{\beta} = \beta_1^2 + 2\beta_1\beta_2 + 3\beta_2^2$ directly.

Corollary 0.3 (Derivative of a simple quadratic form)**.**

$$\underbrace{\frac{\partial}{\partial \beta}(\underset{\sim}{\beta} \cdot \underset{\sim}{\beta})}_{p\times 1} = \underset{p\times 1}{2\underset{\sim}{\beta}}$$

Proof

Proof. Applying Corollary 0.2 with $\mathbf{S} = \mathbf{I}_{p\times p}$ (which is symmetric and constant with respect to $\underset{\sim}{\beta}$):

$$\begin{aligned}
\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}\cdot\underset{\sim}{\beta}) &= \frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}^\top \mathbf{I}_{p\times p}\underset{\sim}{\beta}) && \text{(rewrite with identity matrix)}\\
&= 2\mathbf{I}_{p\times p}\underset{\sim}{\beta} && \text{(derivative of a quadratic form, with } \mathbf{S}=\mathbf{I}_{p\times p})\\
&= 2\underset{\sim}{\beta} && \text{(identity matrix property)}
\end{aligned}$$

□

i Remark (Like the derivative of x^2)

Remark 0.9 (Like the derivative of x^2). This vector derivative is like taking the derivative of x^2: in scalar calculus $\frac{\partial}{\partial x}x^2 = 2x$, and Corollary 0.3 says $\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}\cdot\underset{\sim}{\beta}) = 2\underset{\sim}{\beta}$. For example, with $p=1$ and $\underset{\sim}{\beta} = (\beta_1)$, $\underset{\sim}{\beta}\cdot\underset{\sim}{\beta} = \beta_1^2$, and its derivative is $2\beta_1 = 2\underset{\sim}{\beta}$.

Example 0.20 (Derivative of a sum of squares)**.** Let $\underset{\sim}{\beta} = (\beta_1, \beta_2)^\top$, so $\underset{\sim}{\beta}\cdot\underset{\sim}{\beta} = \beta_1^2 + \beta_2^2$. By Corollary 0.3:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\beta}\cdot\underset{\sim}{\beta})}_{2\times 1} &= 2\underset{\sim}{\beta}\\
&= \begin{pmatrix} 2\beta_1 \\ 2\beta_2 \end{pmatrix}
\end{aligned}$$

Direct partial differentiation yields the same column vector.

Definition 0.11 (Residual vector and residuals)**.** Let $\mathbf{X}$ be an $n\times p$ matrix, $\underset{\sim}{y}\in\mathbb{R}^n$, and $\underset{\sim}{\beta}\in\mathbb{R}^p$. The **residual vector** of $\underset{\sim}{\beta}$ is $\underset{\sim}{\varepsilon}(\underset{\sim}{\beta}) \overset{\text{def}}{=} \underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}$, and its entries $\varepsilon_i = y_i - (\mathbf{X}\underset{\sim}{\beta})_i$ are the **residuals** [11].

Definition 0.12 (Squared error)**.** In the setting of Definition 0.11, the square ε_i^2 is the ith **squared error**.

Definition 0.13 (Residual sum of squares)**.** In the setting of Definition 0.11, the sum of the squared errors (Definition 0.12)

$$\begin{aligned}
\underset{\sim}{\varepsilon}\cdot\underset{\sim}{\varepsilon} &= \sum_{i=1}^{n}\underset{\sim}{\varepsilon}_i^2\\
&= \|\underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}\|^2
\end{aligned}$$

is the **residual sum of squares** [12].

Remark (Least squares solutions minimize the residual sum of squares)

Remark 0.10 (Least squares solutions minimize the residual sum of squares). A least squares solution of $\mathbf{X}\beta = y$ makes $\|y - \mathbf{X}\beta\|$ as small as possible, and so makes the residual sum of squares (Definition 0.13) as small as possible too.

Example 0.21 (Residuals of a constant fit)**.** Let $\mathbf{X} = (1,1)^\top$ ($n = 2$, $p = 1$) and $y = (1,3)^\top$. For $\beta = (2)$, $\mathbf{X}\beta = (2,2)^\top$, so

$$\begin{aligned} \varepsilon(\beta) &= (1,3)^\top - (2,2)^\top && \text{(definition of the residual vector)} \\ &= (-1,1)^\top, && \text{(subtract entry by entry)} \end{aligned}$$

the squared errors are $(-1)^2 = 1$ and $1^2 = 1$, and the residual sum of squares is $1+1 = 2$. For $\beta = (1)$ the residuals are 0 and 2, and the residual sum of squares is $0 + 4 = 4$.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on gradients of sums and sample means of squared errors (Definition 0.13):

- The Gradient of Quadratic Cost
- The Gradient of Mean Squared Error

Theorem 0.12 (Vector chain rule)**.** *Let x be a $p \times 1$ vector, let $y = g(x)$ be a $q \times 1$ vector-valued function of x, and let $z = f(y)$ be a scalar-valued function of y, where g and f have continuous partial derivatives. Then $z = f(g(x))$, as a function of x, satisfies*

$$\underset{p\times 1}{\frac{\partial z}{\partial x}} = \underset{p\times q}{\frac{\partial y}{\partial x}} \underset{q\times 1}{\frac{\partial z}{\partial y}}$$

where $\frac{\partial y}{\partial x}$ is the derivative of Definition 0.6 and $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are vector derivatives (Definition 0.2).

Remark (The order of the factors matters)

Remark 0.11 (The order of the factors matters). The vector chain rule (Theorem 0.12) is like the univariate chain rule, but the order matters now: $\frac{\partial y}{\partial x}$ is $p \times q$ and $\frac{\partial z}{\partial y}$ is $q \times 1$, so the product $\frac{\partial z}{\partial y}\frac{\partial y}{\partial x}$ in the other order is not even defined unless $p = 1$.
The version presented here is for the gradient (Definition 0.2), a column vector. The total

derivative (Definition 0.5), a row vector, is the transpose of the gradient (Theorem 0.5), and transposing both sides gives $\left(\frac{\partial z}{\partial \underset{\sim}{x}}\right)^\top = \left(\frac{\partial z}{\partial \underset{\sim}{y}}\right)^\top \left(\frac{\partial \underset{\sim}{y}}{\partial \underset{\sim}{x}}\right)^\top$, with the factors in the reverse order; there $\left(\frac{\partial \underset{\sim}{y}}{\partial \underset{\sim}{x}}\right)^\top$ is the Jacobian matrix of $\underset{\sim}{y}$ as a function of $\underset{\sim}{x}$ (Definition 0.8).

Example 0.22 (Applying the vector chain rule)**.** Let $\underset{\sim}{x} = (x_1, x_2)^\top$,

$$\begin{aligned} \underset{\sim}{y} &= \underset{\sim}{g}(\underset{\sim}{x}) \\ &= (x_1 + x_2,\ x_1 x_2)^\top, \end{aligned}$$

and

$$\begin{aligned} z &= \underset{\sim}{f}(y) \\ &= y_1^2 + y_2. \end{aligned}$$

Then

$$\begin{aligned} \underset{2\times 1}{\frac{\partial z}{\partial \underset{\sim}{x}}} &= \underset{2\times 2}{\frac{\partial \underset{\sim}{y}}{\partial \underset{\sim}{x}}} \underset{2\times 1}{\frac{\partial z}{\partial \underset{\sim}{y}}} && \text{(vector chain rule)} \\ &= \begin{bmatrix} 1 & x_2 \\ 1 & x_1 \end{bmatrix} \begin{bmatrix} 2y_1 \\ 1 \end{bmatrix} && \text{(differentiate } \underset{\sim}{g} \text{ and } f) \\ &= \begin{bmatrix} 2(x_1 + x_2) + x_2 \\ 2(x_1 + x_2) + x_1 \end{bmatrix} && \text{(multiply, and substitute } y_1 = x_1 + x_2) \\ &= \begin{bmatrix} 2x_1 + 3x_2 \\ 3x_1 + 2x_2 \end{bmatrix} && \text{(collect terms)} \end{aligned}$$

This matches differentiating z directly:

$$\begin{aligned} z &= (x_1 + x_2)^2 + x_1 x_2 \\ &= x_1^2 + 3x_1 x_2 + x_2^2, \end{aligned}$$

so $\frac{\partial}{\partial x_1} z = 2x_1 + 3x_2$ and $\frac{\partial}{\partial x_2} z = 3x_1 + 2x_2$. The product in the other order, a 2×1 matrix times a 2×2 matrix, is not defined. The total derivative is the transpose, the row vector $(2x_1 + 3x_2,\ 3x_1 + 2x_2)$.

See Felippa [13], and Wikipedia contributors [4], section "Relationship with Frechet derivative".

Corollary 0.4 (Vector chain rule for quadratic forms)**.** *If $\underset{\sim}{\varepsilon} = \underset{\sim}{\varepsilon}(\beta)$ is an $n \times 1$ vector-valued function of the $p \times 1$ vector $\underset{\sim}{\beta}$, with continuous partial derivatives, then*

$$\underbrace{\frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\varepsilon}(\underset{\sim}{\beta}) \cdot \underset{\sim}{\varepsilon}(\underset{\sim}{\beta}))}_{p\times 1} = \underbrace{\left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\varepsilon}(\underset{\sim}{\beta})\right)}_{p\times n} \underbrace{(2\underset{\sim}{\varepsilon}(\underset{\sim}{\beta}))}_{n\times 1}$$

Proof

Proof. Apply Theorem 0.12 with $\underset{\sim}{x} = \underset{\sim}{\beta}$, $\underset{\sim}{y} = \underset{\sim}{\varepsilon}$, $q = n$, and

$$\begin{aligned} z &= f(\underset{\sim}{\varepsilon}) \\ &= \underset{\sim}{\varepsilon} \cdot \underset{\sim}{\varepsilon} \\ &= \underset{\sim}{\varepsilon}^\top \underset{\sim}{\varepsilon}\,: \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\varepsilon} \cdot \underset{\sim}{\varepsilon}) &= \left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\varepsilon}\right)\frac{\partial}{\partial \underset{\sim}{\varepsilon}}(\underset{\sim}{\varepsilon}^\top \underset{\sim}{\varepsilon}) && \text{(vector chain rule)} \\ &= \left(\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\varepsilon}\right)(2\underset{\sim}{\varepsilon}) && \text{(derivative of a simple quadratic form, in } \underset{\sim}{\varepsilon}) \end{aligned}$$

The second step is Corollary 0.3, applied to the $n \times 1$ vector $\underset{\sim}{\varepsilon}$ in place of $\underset{\sim}{\beta}$. □

Example 0.23 (Derivative of the residual sum of squares)**.** Let y $(n\times 1)$ and $\mathbf{X}$ $(n\times p)$ be constant with respect to β, and let $\underset{\sim}{\varepsilon}(\underset{\sim}{\beta}) = \underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}$ be the residual vector (Definition 0.11). By Theorem 0.8, $\frac{\partial}{\partial \underset{\sim}{\beta}}(\mathbf{X}\underset{\sim}{\beta}) = \mathbf{X}^\top$, and $\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{y} = \underset{\sim}{\mathbf{0}}_{p\times n}$ because $\underset{\sim}{y}$ is constant, so $\frac{\partial}{\partial \underset{\sim}{\beta}}\underset{\sim}{\varepsilon} = -\mathbf{X}^\top$ $(p \times n)$. By Corollary 0.4:

$$\begin{aligned} \frac{\partial}{\partial \underset{\sim}{\beta}}(\underset{\sim}{\varepsilon} \cdot \underset{\sim}{\varepsilon}) &= (-\mathbf{X}^\top)(2\underset{\sim}{\varepsilon}) && \text{(vector chain rule for quadratic forms)} \\ &= -2\,\mathbf{X}^\top(\underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}) && \text{(substitute } \underset{\sim}{\varepsilon} = \underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}) \end{aligned}$$

Setting this $p \times 1$ vector to $\underset{\sim}{0}$ gives the normal equations $\mathbf{X}^\top\mathbf{X}\underset{\sim}{\beta} = \mathbf{X}^\top\underset{\sim}{y}$ of least squares.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on the chain rule for partial derivatives:

- The Chain Rule for Partial Derivatives
- Exercises on the Multivariate Chain Rule

Definition 0.14 (Matrix derivative (gradient))**.** For a scalar-valued function $f(\mathbf{X})$ of an $m \times n$ matrix $\mathbf{X}$, the **matrix derivative** [9] is the $m \times n$ matrix whose (i, j) entry is the partial derivative of f with respect to the (i, j) entry of $\mathbf{X}$:

$$\left[\frac{\partial}{\partial \mathbf{X}} f\right]_{ij} = \frac{\partial}{\partial X_{ij}} f$$

Like the vector derivative (Definition 0.2), the matrix derivative is also called the **gradient** of f with respect to $\mathbf{X}$, written $\nabla_{\mathbf{X}} f$.

Example 0.24 (The matrix derivative of a trace)**.** Let $\mathbf{X}$ be a 2×2 matrix and

$$\begin{aligned} f(\mathbf{X}) &= \operatorname{tr}(\mathbf{X}) \\ &= X_{11} + X_{22} \end{aligned}$$

(see trace). Then $\frac{\partial}{\partial X_{ij}} f = 1$ if $i = j$ and 0 otherwise, so:

$$\frac{\partial}{\partial \mathbf{X}} f = \mathbf{I}_2$$

Theorem 0.13 (Matrix derivative of the trace of a matrix product)**.** *If* $\mathbf{A}$ *$(r \times m)$ and* $\mathbf{B}$ *$(n \times r)$ are constant with respect to the $m \times n$ matrix* $\mathbf{X}$*, then:*

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{AXB})}_{m \times n} = \underbrace{\mathbf{A}^\top}_{m \times r} \underbrace{\mathbf{B}^\top}_{r \times n}$$

Proof

Proof. Write A_{kl}, X_{kl}, and B_{kl} for the entries of $\mathbf{A}$, $\mathbf{X}$, and $\mathbf{B}$. For entry (i, j):

$$\begin{aligned} \left[\frac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{AXB})\right]_{ij} &= \frac{\partial}{\partial X_{ij}} \sum_{a=1}^{r} \sum_{b=1}^{m} \sum_{c=1}^{n} A_{ab} X_{bc} B_{ca} && \text{(trace of the } r \times r \text{ product } \mathbf{AXB}) \\ &= \sum_{a=1}^{r} A_{ai} B_{ja} && \text{(only the terms with } b = i,\ c = j \text{ depend on } X_{ij}) \\ &= \sum_{a=1}^{r} [\mathbf{A}^\top]_{ia} [\mathbf{B}^\top]_{aj} && \text{(definition of the transpose)} \\ &= [\mathbf{A}^\top \mathbf{B}^\top]_{ij} && \text{(definition of matrix multiplication)} \end{aligned}$$

□

Remark (Why the theorem uses the trace)

Remark 0.12 (Why the theorem uses the trace). The trace makes $\operatorname{tr}(\mathbf{AXB})$ a scalar, so its matrix derivative (Definition 0.14) is again an $m \times n$ matrix. The matrix product $\mathbf{AXB}$ itself (without the trace) is an $r \times r$ matrix, and each of its r^2 entries has a partial derivative with respect to each of the mn entries of $\mathbf{X}$. Those r^2mn partial derivatives form a four-index array (a fourth-order tensor), not a matrix, which is why this result is stated for the scalar $\operatorname{tr}(\mathbf{AXB})$.
For example, with

$$\begin{aligned}\mathbf{A} &= \mathbf{B} \\ &= \mathbf{I}_2,\end{aligned}$$

the product $\mathbf{AXB} = \mathbf{X}$ has 4 entries, each with 4 partial derivatives, 16 in all, while its trace $X_{11}+X_{22}$ has the 4 partial derivatives that form the 2×2 matrix $\mathbf{I}_2$ of Example 0.24.

Example 0.25 (Differentiating a weighted trace)**.** Let $\mathbf{A} = \mathbf{I}_2$ (2×2) and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2). Then $\operatorname{tr}(\mathbf{AXB}) = 2X_{11} + 3X_{22}$, and:

$$\begin{aligned}\underbrace{\frac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{AXB})}_{2\times2} &= \underbrace{\mathbf{A}^\top}_{2\times2}\,\underbrace{\mathbf{B}^\top}_{2\times2} \\ &= \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\end{aligned}$$

Exercise 0.3 (A gradient that does not mention its variable)**.** Take

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix}$$

let $W \in \mathbb{R}^{2\times3}$ vary, and define

$$g(W) = \sum_{i=1}^{2} \sum_{j=1}^{3} W_{ij} A_{ij}$$

which is the entry-by-entry version of the inner product for matrices. Compute $\nabla_W g(W)$, and say what is unusual about the answer.

Solution

Solution 0.3. Written out over the six positions,

$$g(W) = 2W_{11} - W_{12} + 0\,W_{13} + 4W_{21} + 3W_{22} - 2W_{23}$$

Each W_{ij} appears in exactly one term, multiplied by A_{ij} and by nothing else, so differentiating with respect to it leaves A_{ij} behind:

$$\frac{\partial g}{\partial W_{ij}} = A_{ij}$$

Collecting those partial derivatives into a matrix of the same shape as W, as Table 1 requires,

$$\begin{aligned}\nabla_W g(W) &= \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix} \\ &= A\end{aligned}$$

What is unusual is that the gradient came out **constant**: it does not mention W at all. The constant answer is not a quirk of this particular A. g is a linear function of W, and the gradient of a linear function is constant everywhere, for the same reason the derivative of $f(w) = cw$ is c no matter where it is evaluated.
A constant gradient is the easy case, and it is not the case we usually face. Most functions we minimize are curved — the residual sum of squares of a linear model (Definition 0.13) is the standard example — so their gradient changes from point to point and the downhill direction has to be worked out afresh at every step.

Theorem 0.14 (Gradient of an entrywise inner product with a fixed matrix)**.** *Let A be a fixed $m \times n$ matrix, and for $W \in \mathbb{R}^{m \times n}$ define*

$$g(W) = \sum_{i=1}^{m} \sum_{j=1}^{n} W_{ij} A_{ij}$$

Then for every W, $\nabla_W g(W) = A$. In particular, the gradient is constant: it does not depend on W.

Proof

Proof. This is the solution to Exercise 0.3, worked in Solution 0.3, where $m = 2$, $n = 3$, and the entries of A are given. The same argument works for every A: each W_{ij} appears in exactly one term, multiplied by A_{ij}, so $\frac{\partial g}{\partial W_{ij}} = A_{ij}$. □

Second partial derivatives and the Hessian

Definition 0.1 (Open ball)**.** Let $\underset{\sim}{x} \in \mathbb{R}^p$ and $r > 0$. The **open ball** [14] of radius r around $\underset{\sim}{x}$ is the set $\{\underset{\sim}{y} \in \mathbb{R}^p : \|\underset{\sim}{y} - \underset{\sim}{x}\| < r\}$, of points whose distance from $\underset{\sim}{x}$ (the Euclidean norm of the difference) is less than r.

Definition 0.2 (Closed ball)**.** For $\underset{\sim}{x}$ and r as in Definition 0.1, the **closed ball** of radius r around $\underset{\sim}{x}$ is $\{\underset{\sim}{y} \in \mathbb{R}^p : \|\underset{\sim}{y} - \underset{\sim}{x}\| \leq r\}$.

Definition 0.3 (Unit ball)**.** The open and closed balls (Definition 0.1, Definition 0.2) of radius 1 around $\underset{\sim}{0}$ are the open and closed **unit balls**.

Example 0.1 (Points in the unit balls of $\mathbb{R}^2$)**.**

- $(0.5, 0.5)^\top$ has

$$\begin{aligned} \left\|(0.5, 0.5)^\top\right\|^2 &= 0.25 + 0.25 \\ &= 0.5 \\ &< 1, \end{aligned}$$

  so it is in both the open and the closed unit ball.

- $(0.6, 0.8)^\top$ has

$$\begin{aligned} \left\|(0.6, 0.8)^\top\right\|^2 &= 0.36 + 0.64 \\ &= 1, \end{aligned}$$

  so it is in the closed unit ball but not the open one.

- For $p = 1$, the open ball of radius r around c is the open interval $(c - r, c + r)$.

Definition 0.4 (Second partial derivative)**.** Let f be a scalar-valued function of a $p \times 1$ vector $\underset{\sim}{x}$, and let $i, j \in \{1, \ldots, p\}$. Suppose the partial derivative $\frac{\partial}{\partial x_j} f$ (Definition 0.1) exists at every point of an open ball around $\underset{\sim}{x}$ (Definition 0.1). The **second partial derivative** of f at $\underset{\sim}{x}$, first with respect to x_j and then with respect to x_i, is

$$\frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right),$$

the partial derivative with respect to x_i of the function $\frac{\partial}{\partial x_j} f$, when it exists.

Example 0.2 (Second partial derivatives of $x_1^2 x_2 + 3x_2$)**.** For $f(\underset{\sim}{x}) = x_1^2 x_2 + 3x_2$ of Example 0.1, $\frac{\partial}{\partial x_1} f(\underset{\sim}{x}) = 2x_1 x_2$ and $\frac{\partial}{\partial x_2} f(\underset{\sim}{x}) = x_1^2 + 3$. Differentiating each of these again:

- $$\begin{aligned} \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_1} f(\underset{\sim}{x})\right) &= \frac{\partial}{\partial x_1}(2x_1 x_2) \\ &= 2x_2; \end{aligned}$$
- $$\begin{aligned} \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1} f(\underset{\sim}{x})\right) &= \frac{\partial}{\partial x_2}(2x_1 x_2) \\ &= 2x_1; \end{aligned}$$
- $$\begin{aligned} \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2} f(\underset{\sim}{x})\right) &= \frac{\partial}{\partial x_1}(x_1^2 + 3) \\ &= 2x_1; \end{aligned}$$
- $$\begin{aligned} \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_2} f(\underset{\sim}{x})\right) &= \frac{\partial}{\partial x_2}(x_1^2 + 3) \\ &= 0. \end{aligned}$$

The two mixed partial derivatives are both $2x_1$.

Definition 0.5 (Mixed partial derivative)**.** When $i \neq j$, the second partial derivative of Definition 0.4 is a **mixed partial derivative**.

Definition 0.6 (Hessian matrix)**.** Let f be a scalar-valued function of a $p \times 1$ vector $\underset{\sim}{x}$ whose first partial derivatives exist on an open ball around $\underset{\sim}{x}$ (Definition 0.1) and whose second partial derivatives (Definition 0.4) exist at $\underset{\sim}{x}$. The **Hessian matrix** [15] of f at $\underset{\sim}{x}$ is the derivative (Definition 0.6) of the transposed gradient $\left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})\right)^\top$ (Definition 0.2):

$$\underbrace{\mathbf{H}_f(\underset{\sim}{x})}_{p \times p} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}}\left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})\right)^\top. \tag{5}$$

Example 0.3 (A Hessian matrix)**.** Let $f(\underset{\sim}{x}) = e^{2x_1+x_2} - x_1$ for $\underset{\sim}{x} = (x_1, x_2)^\top$, where e is Euler's number, and write $u = 2x_1 + x_2$. By the chain rule, $\frac{\partial}{\partial x_1} e^u = 2e^u$ and $\frac{\partial}{\partial x_2} e^u = e^u$, so

$$\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \begin{bmatrix} 2e^u - 1 \\ e^u \end{bmatrix}.$$

Differentiating each entry again,

$$\begin{aligned} \mathbf{H}_f(\underset{\sim}{x}) &= \begin{bmatrix} \frac{\partial}{\partial x_1}(2e^u - 1) & \frac{\partial}{\partial x_1} e^u \\ \frac{\partial}{\partial x_2}(2e^u - 1) & \frac{\partial}{\partial x_2} e^u \end{bmatrix} \\ &= \begin{bmatrix} 4e^u & 2e^u \\ 2e^u & e^u \end{bmatrix}, \end{aligned}$$

and at $\underset{\sim}{x} = \underset{\sim}{0}$, where $u = 0$, $\mathbf{H}_f(\underset{\sim}{0}) = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$.

Example 0.4 (A function with a gradient but no Hessian at a point)**.** Let $f(\underset{\sim}{x}) = x_1|x_1|$ for $\underset{\sim}{x} = (x_1, x_2)^\top$, where $|\cdot|$ is the absolute value. For $x_1 > 0$, $f = x_1^2$ and $\frac{\partial}{\partial x_1} f = 2x_1$; for $x_1 < 0$, $f = -x_1^2$ and $\frac{\partial}{\partial x_1} f = -2x_1$; and at $x_1 = 0$ the difference quotient is $h|h|/h = |h| \to 0$. So the gradient exists everywhere: it is $(2|x_1|,\ 0)^\top$. But $2|x_1|$ has no derivative in x_1 at $x_1 = 0$ (its difference quotient $2|h|/h$ is 2 for $h > 0$ and -2 for $h < 0$), so $\left[\mathbf{H}_f(\underset{\sim}{x})\right]_{11}$, and with it the Hessian, does not exist at any $\underset{\sim}{x}$ with $x_1 = 0$.

Theorem 0.1 (Other ways to write the Hessian, and its entries)**.** *Let f and $\underset{\sim}{x}$ be as in Definition 0.6.*

1. *The Hessian can be written with the row-vector derivative (Definition 0.5) of f:*

$$\begin{aligned} \mathbf{H}_f(\underset{\sim}{x}) &= \frac{\partial}{\partial \underset{\sim}{x}} \left(\frac{\partial f(\underset{\sim}{x})}{\partial \underset{\sim}{x}^\top} \right) \\ &= \frac{\partial^2 f(\underset{\sim}{x})}{\partial \underset{\sim}{x} \partial \underset{\sim}{x}^\top}. \end{aligned} \tag{6}$$

2. *The (i, j) entry of the Hessian is*

$$\left[\mathbf{H}_f(\underset{\sim}{x})\right]_{ij} = \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x}) \right). \tag{7}$$

Proof

Proof. **Part 1.** By Theorem 0.5, the transposed gradient $\left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})\right)^\top$ equals the row-vector derivative $\frac{\partial f(\underset{\sim}{x})}{\partial \underset{\sim}{x}^\top}$. Substituting it into Equation 5 gives the first line. The second line is the same expression written as a second derivative. It follows the shape rule of matrix multiplication (Remark 0.4): the $p \times 1$ column-vector operator $\frac{\partial}{\partial \underset{\sim}{x}}$ stands on the left of a $1 \times p$ row vector, giving a $p \times p$ matrix.

Part 2. Let $y_j \overset{\text{def}}{=} \frac{\partial}{\partial x_j} f(\underset{\sim}{x})$ be entry j of the gradient. By Definition 0.6, the (i, j) entry of the derivative of the vector $\underset{\sim}{y} = (y_1, \dots, y_p)$ with respect to $\underset{\sim}{x}$ is

$$\begin{aligned}
\left[\mathbf{H}_f(\underset{\sim}{x})\right]_{ij} &= \left[\frac{\partial}{\partial \underset{\sim}{x}} \underset{\sim}{y}^\top\right]_{ij} && \text{(Equation 5, with } \underset{\sim}{y} \text{ the gradient)} \\
&= \frac{\partial}{\partial x_i} y_j && \text{(Definition 0.6)} \\
&= \frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right). && \left(y_j = \frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right)
\end{aligned}$$

□

Example 0.5 (Computing the Hessian entry by entry, and checking it by finite differences)**.** Take $f(\underset{\sim}{x}) = e^{2x_1 + x_2} - x_1$ from Example 0.3. By Equation 7, entry (i, j) of the Hessian is the derivative with respect to x_i of the derivative with respect to x_j. The code differentiates `f` symbolically in that order with `D()`, evaluates the four entries at $\underset{\sim}{x} = (0.5, -1)$, and compares them with central differences of the gradient.

```r
f_expr <- quote(exp(2 * x1 + x2) - x1)
vars <- c("x1", "x2")
point <- c(x1 = 0.5, x2 = -1)

grad_expr <- lapply(vars, function(v) D(f_expr, v))
hess_expr <- lapply(vars, function(i) {
  lapply(grad_expr, function(g_j) D(g_j, i))
})
H <- sapply(1:2, function(j) {
  sapply(1:2, function(i) eval(hess_expr[[i]][[j]], as.list(point)))
})

grad_at <- function(x) {
  sapply(grad_expr, function(g) eval(g, as.list(setNames(x, vars))))
}
h <- 1e-5
H_fd <- sapply(1:2, function(j) {
  sapply(1:2, function(i) {
    step <- replace(c(0, 0), i, h)
    (grad_at(point + step)[j] - grad_at(point - step)[j]) / (2 * h)
  })
})

H
#>      [,1] [,2]
#> [1,]    4    2
#> [2,]    2    1
H_fd
#>      [,1] [,2]
#> [1,]    4    2
#> [2,]    2    1
```

At this point $u = 2x_1 + x_2 = 0$, so the closed form in Example 0.3 gives $4e^u = 4$ for entry $(1, 1)$, and the code gives 4. The off-diagonal entries are 2 and 2, which are equal. The finite differences agree with the symbolic entries to within 0.00000000026.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on second and higher partial derivatives:

- Higher-Order Partial Derivatives
- Exercise on Higher-Order Partial Derivatives

Definition 0.7 (Continuity at a point in several variables)**.** A function f from $\mathbb{R}^p$ to $\mathbb{R}^q$ is **continuous at** $\underset{\sim}{x}$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\|f(\underset{\sim}{y}) - f(\underset{\sim}{x})\| < \varepsilon$ whenever $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta$. For $q = 1$, $\|f(\underset{\sim}{y}) - f(\underset{\sim}{x})\| = |f(\underset{\sim}{y}) - f(\underset{\sim}{x})|$.

Definition 0.8 (Continuity on a set in several variables)**.** A function f from $\mathbb{R}^p$ to $\mathbb{R}^q$ is **continuous on** a set if it is continuous at every point of the set (Definition 0.7).

Example 0.6 (A continuous function, and a discontinuous one)**.**

- $f(\underset{\sim}{x}) = x_1 + x_2$ is continuous at every $\underset{\sim}{x}$. Each $|y_i - x_i| \le \|\underset{\sim}{y} - \underset{\sim}{x}\|$, since the squared length is a sum of nonnegative squares, so

$$\begin{aligned} |f(\underset{\sim}{y}) - f(\underset{\sim}{x})| &= |(y_1 - x_1) + (y_2 - x_2)| && \text{(subtract)} \\ &\le |y_1 - x_1| + |y_2 - x_2| && \text{(triangle inequality for numbers)} \\ &\le 2\,\|\underset{\sim}{y} - \underset{\sim}{x}\|, && \text{(each term is at most } \|\underset{\sim}{y} - \underset{\sim}{x}\|) \end{aligned}$$

  which is less than ε whenever $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta = \varepsilon/2$.

- $f(\underset{\sim}{x}) = 1$ if $x_1 > 0$ and $f(\underset{\sim}{x}) = 0$ otherwise is not continuous at $\underset{\sim}{0}$: for $\varepsilon = \frac{1}{2}$ and any $\delta > 0$, the point $\underset{\sim}{y} = (\delta/2, 0)^\top$ has $\|\underset{\sim}{y} - \underset{\sim}{0}\| = \delta/2 < \delta$ but $|f(\underset{\sim}{y}) - f(\underset{\sim}{0})| = 1 > \frac{1}{2}$.

Theorem 0.2 (Continuity in one variable, and sums, multiples and compositions)**.**

1. *For $p = q = 1$, a function f is continuous at x in the sense of Definition 0.7 exactly when $\lim_{y \to x} f(y) = f(x)$ (limit), which is the usual continuity at x.*
2. *If $f : \mathbb{R}^p \to \mathbb{R}^q$ is continuous at $\underset{\sim}{x}$ and $g : \mathbb{R}^q \to \mathbb{R}^r$ is continuous at $f(\underset{\sim}{x})$, then the composition $g \circ f$ is continuous at $\underset{\sim}{x}$.*
3. *If $f, g : \mathbb{R}^p \to \mathbb{R}$ are continuous at $\underset{\sim}{x}$, then $f + g$ is continuous at $\underset{\sim}{x}$.*
4. *If $f : \mathbb{R}^p \to \mathbb{R}$ is continuous at $\underset{\sim}{x}$ and $c \in \mathbb{R}$, then cf is continuous at $\underset{\sim}{x}$.*

If the hypotheses of part 2, 3 or 4 hold at every point of a set, the conclusion holds at every point of that set.

Proof

Proof. Let $\varepsilon > 0$ be given.
Part 1. For $p = q = 1$, $\|f(y) - f(x)\| = |f(y) - f(x)|$ and $\|y - x\| = |y - x|$. Definition 0.7 then asks for a $\delta > 0$ such that $|f(y) - f(x)| < \varepsilon$ whenever $|y - x| < \delta$. The limit definition with $L = f(x)$ asks for the same inequality, but only whenever $0 < |y - x| < \delta$. The two agree, because at $y = x$ the inequality $|f(x) - f(x)| = 0 < \varepsilon$ holds for every δ.
Part 2. Because g is continuous at $f(\underset{\sim}{x})$, there is a $\delta_g > 0$ such that $\|g(\underset{\sim}{z}) - g(f(\underset{\sim}{x}))\| < \varepsilon$ whenever $\|\underset{\sim}{z} - f(\underset{\sim}{x})\| < \delta_g$. Because f is continuous at $\underset{\sim}{x}$, applied with δ_g in the role of ε, there is a $\delta > 0$ such that $\|f(\underset{\sim}{y}) - f(\underset{\sim}{x})\| < \delta_g$ whenever $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta$. So if $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta$, then $\underset{\sim}{z} = f(\underset{\sim}{y})$ has $\|\underset{\sim}{z} - f(\underset{\sim}{x})\| < \delta_g$, and therefore $\|g(f(\underset{\sim}{y})) - g(f(\underset{\sim}{x}))\| < \varepsilon$.
Part 3. Because f and g are continuous at $\underset{\sim}{x}$, there are $\delta_f > 0$ and $\delta_g > 0$ such that $|f(\underset{\sim}{y}) - f(\underset{\sim}{x})| < \varepsilon/2$ whenever $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta_f$, and $|g(\underset{\sim}{y}) - g(\underset{\sim}{x})| < \varepsilon/2$ whenever $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta_g$. Let $\delta \stackrel{\text{def}}{=} \min\{\delta_f, \delta_g\}$. If $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta$, then

$$\begin{aligned}
|(f+g)(\underset{\sim}{y}) - (f+g)(\underset{\sim}{x})| &= |(f(\underset{\sim}{y}) - f(\underset{\sim}{x})) + (g(\underset{\sim}{y}) - g(\underset{\sim}{x}))| && \text{(regroup)} \\
&\le |f(\underset{\sim}{y}) - f(\underset{\sim}{x})| + |g(\underset{\sim}{y}) - g(\underset{\sim}{x})| && \text{(triangle inequality for numbers)} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} && \text{(choice of } \delta) \\
&= \varepsilon. && \text{(add)}
\end{aligned}$$

Part 4. Because f is continuous at $\underset{\sim}{x}$, there is a $\delta > 0$ such that $|f(\underset{\sim}{y}) - f(\underset{\sim}{x})| < \varepsilon/(|c|+1)$ whenever $\|\underset{\sim}{y} - \underset{\sim}{x}\| < \delta$. For such $\underset{\sim}{y}$,

$$\begin{aligned}
|cf(\underset{\sim}{y}) - cf(\underset{\sim}{x})| &= |c|\,|f(\underset{\sim}{y}) - f(\underset{\sim}{x})| && \text{(factor out } c) \\
&\le |c|\,\frac{\varepsilon}{|c|+1} && \text{(choice of } \delta) \\
&< \varepsilon. && (|c| < |c|+1)
\end{aligned}$$

□

Example 0.7 (Choosing δ for a sum, a multiple and a composition)**.** Let $f(\underset{\sim}{x}) = x_1$ and $g(\underset{\sim}{x}) = x_2$ on $\mathbb{R}^2$, and $h(t) = 3t$ on $\mathbb{R}$. Since $|y_i - x_i| \le \|\underset{\sim}{y} - \underset{\sim}{x}\|$, the functions f and g are continuous with $\delta = \varepsilon$, and h is continuous with $\delta = \varepsilon/3$, because $|h(s) - h(t)| = 3\,|s - t|$. Take $\underset{\sim}{x} = (1, 2)$ and $\varepsilon = 0.1$. The proof of Theorem 0.2 gives these values of δ:

- for $f + g$ (part 3), $\delta = \min\{\varepsilon/2, \varepsilon/2\}$;
- for $3f$ (part 4), $\delta = \varepsilon/(3+1)$;
- for $h \circ (f + g)$ (part 2), the outer value $\delta_h = \varepsilon/3$ plays the role of ε for $f + g$, so $\delta = \delta_h/2$.

The code draws many points $\underset{\sim}{y}$ at distance less than δ from $\underset{\sim}{x}$ and finds the largest change

in each function.

```
set.seed(571)
eps <- 0.1
x <- c(1, 2)
f <- function(x) x[1]
g <- function(x) x[2]
h <- function(t) 3 * t

fun_sum <- function(x) f(x) + g(x)
fun_multiple <- function(x) 3 * f(x)
fun_composition <- function(x) h(fun_sum(x))

deltas <- c(
  sum = eps / 2,
  multiple = eps / (3 + 1),
  composition = (eps / 3) / 2
)
funs <- list(
  sum = fun_sum, multiple = fun_multiple, composition = fun_composition
)

n <- 10000L
largest_change <- sapply(names(funs), function(nm) {
  radius <- deltas[[nm]] * sqrt(runif(n))
  angle <- runif(n, 0, 2 * pi)
  y <- cbind(x[1] + radius * cos(angle), x[2] + radius * sin(angle))
  max(abs(apply(y, 1, funs[[nm]]) - funs[[nm]](x)))
})
data.frame(delta = deltas, largest_change = largest_change, eps = eps)
#>                  delta largest_change eps
#> sum         0.05000000     0.07063484 0.1
#> multiple    0.02500000     0.07494539 0.1
#> composition 0.01666667     0.07063336 0.1
```

For 3 of the 3 functions, the largest change among the 10000 points is below $\varepsilon = 0.1$, as Theorem 0.2 guarantees.

Example 0.8 (Partial derivatives can exist where a function is not continuous)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be

$$f(\underset{\sim}{x}) = \begin{cases} \dfrac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } \underset{\sim}{x} \neq \underset{\sim}{0}, \\ 0 & \text{if } \underset{\sim}{x} = \underset{\sim}{0}. \end{cases}$$

Both partial derivatives exist at $\underset{\sim}{0}$. For $h \neq 0$,

$$\begin{aligned} f(h, 0) &= \frac{h \cdot 0}{h^2 + 0} \\ &= 0, \end{aligned}$$

so

$$\begin{aligned} \frac{f(0+h, 0) - f(0,0)}{h} &= \frac{0 - 0}{h} && (f(h,0) = 0 \text{ and } f(0,0) = 0) \\ &= 0, && (0/h = 0 \text{ for } h \neq 0) \end{aligned}$$

and the limit as $h \to 0$ is 0: $\frac{\partial}{\partial x_1} f(\underset{\sim}{0}) = 0$ (Definition 0.1). The same steps with $f(0, h) = 0$ give $\frac{\partial}{\partial x_2} f(\underset{\sim}{0}) = 0$.

But f is not continuous at $\underset{\sim}{0}$ (Definition 0.7). Take $\varepsilon = \frac{1}{4}$ and any $\delta > 0$, and let $t = \delta/2$ and $\underset{\sim}{y} = (t, t)^\top$. Then

$$\begin{aligned} \|\underset{\sim}{y} - \underset{\sim}{0}\| &= \sqrt{t^2 + t^2} \\ &= t\sqrt{2} \\ &= \delta/\sqrt{2} \\ &< \delta, \end{aligned}$$

but

$$\begin{aligned} |f(\underset{\sim}{y}) - f(\underset{\sim}{0})| &= \left| \frac{t \cdot t}{t^2 + t^2} - 0 \right| && (\underset{\sim}{y} \neq \underset{\sim}{0}, \text{ since } t > 0) \\ &= \frac{t^2}{2t^2} && (t^2 + t^2 = 2t^2) \\ &= \tfrac{1}{2}, && (\text{cancel } t^2 > 0) \end{aligned}$$

which is more than $\varepsilon = \frac{1}{4}$.

The partial derivatives only look along the two coordinate axes, where f is 0; along the line $x_1 = x_2$, f is $\frac{1}{2}$ everywhere except at $\underset{\sim}{0}$. So partial derivatives existing at a point says little about how f behaves near it.

Theorem 0.3 (Symmetry of the Hessian)**.** *If the second partial derivatives of f exist and are continuous (Definition 0.7) on an open ball around $\underset{\sim}{x}$ (Definition 0.1), then $\frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right) = \frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_i} f(\underset{\sim}{x})\right)$ for all i, j [16], so $\mathbf{H}_f(\underset{\sim}{x})$ (Definition 0.6) is symmetric*

(symmetric matrix).

The proof applies the one-variable mean value theorem twice, which these notes do not develop; see [17], Theorem 9.41, which is stated for two variables: apply it to f as a function of x_i and x_j, with the other coordinates held fixed.

Example 0.9 (Mixed partial derivatives agree)**.** In Example 0.3, the $(1,2)$ and $(2,1)$ entries of $\mathbf{H}_f(\underset{\sim}{x})$ are both $2e^{2x_1+x_2}$.

Example 0.10 (Without continuity, the mixed partials can differ)**.** Let $f(x_1,x_2) = \dfrac{x_1x_2(x_1^2-x_2^2)}{x_1^2+x_2^2}$ for $\underset{\sim}{x} \neq \underset{\sim}{0}$, and $f(\underset{\sim}{0}) = 0$. For $x_2 \neq 0$, the partial derivative (Definition 0.1) in x_1 at $(0,x_2)$ is

$$\begin{aligned}
\frac{\partial}{\partial x_1} f(0,x_2) &= \lim_{h\to 0} \frac{f(h,x_2)-f(0,x_2)}{h} && \text{(definition of the partial derivative, Definition 0.1)}\\
&= \lim_{h\to 0} \frac{x_2(h^2-x_2^2)}{h^2+x_2^2} && (f(0,x_2)=0\text{; cancel } h)\\
&= \frac{x_2\cdot(-x_2^2)}{x_2^2}\\
&= -x_2, && \text{(the quotient is continuous at } h=0)
\end{aligned}$$

and

$$\begin{aligned}
\frac{\partial}{\partial x_1} f(\underset{\sim}{0}) &= \lim_{h\to 0}(0-0)/h\\
&= 0,
\end{aligned}$$

so $\frac{\partial}{\partial x_1} f(0,x_2) = -x_2$ holds at $x_2 = 0$ too. In the same way, with the roles of x_1 and x_2 swapped, $f(x_1,k)/k = x_1(x_1^2-k^2)/(x_1^2+k^2) \to x_1$, so $\frac{\partial}{\partial x_2} f(x_1,0) = x_1$ for every x_1. So at $\underset{\sim}{0}$ (Definition 0.6)

$$
\begin{aligned}
\left[\mathbf{H}_f(\underset{\sim}{0})\right]_{21} &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1} f\right) \\
&= \frac{d}{dx_2}(-x_2) \\
&= -1, \\
\left[\mathbf{H}_f(\underset{\sim}{0})\right]_{12} &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2} f\right) \\
&= \frac{d}{dx_1} x_1 \\
&= 1 :
\end{aligned}
$$

the Hessian at $\underset{\sim}{0}$ is not symmetric. By Theorem 0.3, then, the second partial derivatives of this f cannot all be continuous near $\underset{\sim}{0}$.

Theorem 0.4 (Hessian of a quadratic form)**.** *If* $\mathbf{S}$ *is a symmetric* $p \times p$ *matrix that is constant with respect to* $\underset{\sim}{x}$*, then* $f(\underset{\sim}{x}) = \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x}$ *has* $\mathbf{H}_f(\underset{\sim}{x}) = 2\mathbf{S}$ *for every* $\underset{\sim}{x}$*.*

Proof

Proof.

$$
\begin{aligned}
\mathbf{H}_f(\underset{\sim}{x}) &= \frac{\partial}{\partial \underset{\sim}{x}}\left(\frac{\partial}{\partial \underset{\sim}{x}}(\underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x})\right)^\top && \text{(Definition 0.6)} \\
&= \frac{\partial}{\partial \underset{\sim}{x}}(2\mathbf{S}\underset{\sim}{x})^\top && \text{(Corollary 0.2)} \\
&= \left(\frac{\partial}{\partial \underset{\sim}{x}} \underset{\sim}{x}\right)(2\mathbf{S})^\top && \text{(Theorem 0.9, with } \mathbf{A} = 2\mathbf{S} \text{ and } \underset{\sim}{v} = \underset{\sim}{x};\ \frac{\partial}{\partial \underset{\sim}{x}}(2\mathbf{S}\underset{\sim}{x})^\top = \frac{\partial}{\partial \underset{\sim}{x}}(2\mathbf{S}\underset{\sim}{x}) \text{, Definition 0.6)} \\
&= \mathbf{I}_p\,(2\mathbf{S})^\top && (\frac{\partial}{\partial \underset{\sim}{x}} \underset{\sim}{x} = \mathbf{I}_p \text{, Remark 0.6)} \\
&= 2\mathbf{S}. && (\mathbf{S} \text{ is symmetric})
\end{aligned}
$$

□

Example 0.11 (The Hessian of a 2×2 quadratic form)**.** For $\mathbf{S} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ as in Example 0.19, Theorem 0.4 gives $\mathbf{H}_f(\underset{\sim}{x}) = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$. Directly, the gradient found there, with β_i renamed x_i, is $(6x_1 + 2x_2,\ 2x_1 + 4x_2)^\top$, and differentiating its entries by x_1 and by x_2 (Definition 0.6) gives

$$\begin{aligned}\mathbf{H}_f(\underset{\sim}{x}) &= \begin{bmatrix} \frac{\partial}{\partial x_1}(6x_1 + 2x_2) & \frac{\partial}{\partial x_1}(2x_1 + 4x_2) \\ \frac{\partial}{\partial x_2}(6x_1 + 2x_2) & \frac{\partial}{\partial x_2}(2x_1 + 4x_2) \end{bmatrix} \\ &= \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}.\end{aligned}$$

Further reading

- Marsden and Tromba [18] is a standard textbook on multivariable and vector calculus. It covers differentiation of functions of several variables, multiple integrals, line and surface integrals, and the theorems of Green, Gauss, and Stokes.
- Petersen and Pedersen [19] is a free desk reference that collects matrix identities. Its chapter on derivatives lists derivatives of vector and matrix expressions, such as the linear and quadratic forms on this page.
- Miller [3] The partial derivative review exercises and worked solutions on this page are adapted from this supplemental review chapter.
- Wikipedia contributors [4] — the gradient

See also the Linear Algebra and Vector Calculus further reading.

- Hua Zhou's lecture notes for "UCLA Biostat 216 - Mathematical Methods for Biostatistics" (2023 Fall)
- Neural Networks is a YouTube playlist by Grant Sanderson (3Blue1Brown); its chapters on gradient descent and backpropagation calculus visually illustrate how gradients of multivariate cost functions are computed and used for optimization.

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